

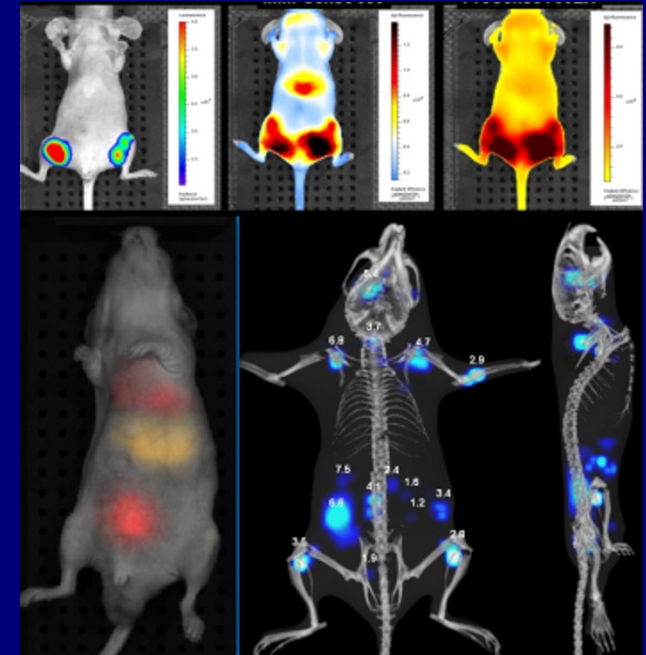
In Vivo Imaging System

IVIS Lumina III

陳韋翰 Tim Chen

連絡電話：0800-898-178

E-mail：support@jnhtech.com.tw



What Will Be Covered?

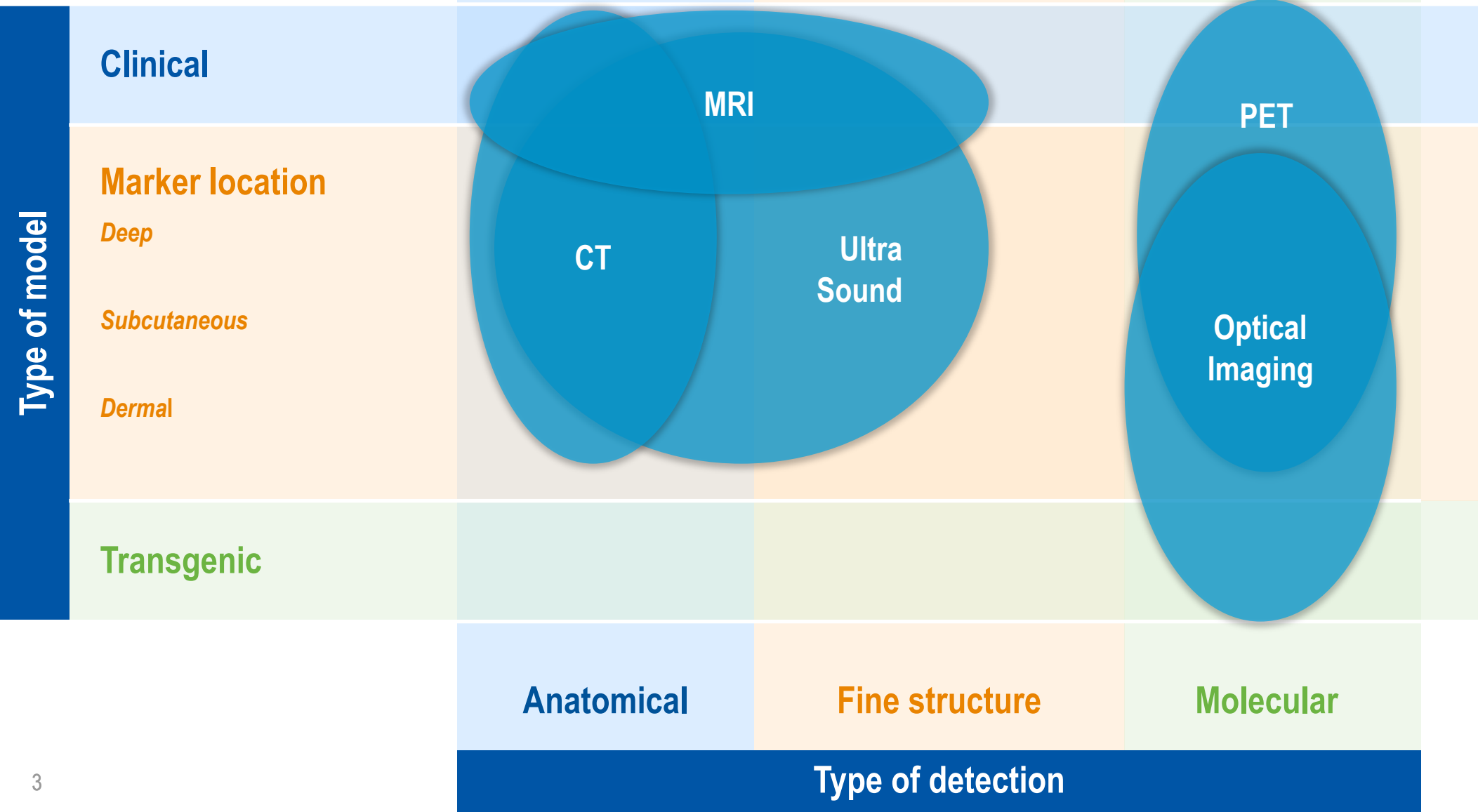
Introduction

- Principles of Optical 'In Vivo' Imaging
- Key IVIS® Hardware components
- Overview of Living Image® Software
- Application

Training

- Hands on Training

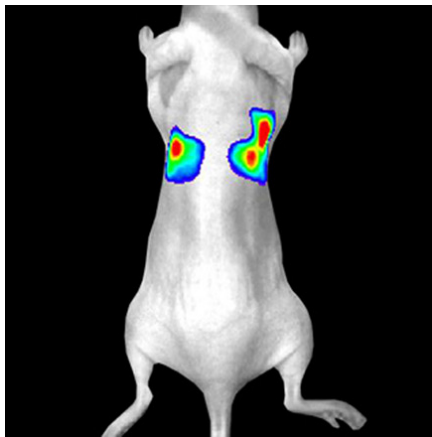
In Vivo Imaging LandScape



- ▶ Gene expression results in production of luciferase or fluorescence protein
 - Amount of light is proportional to number of live active cells
- ▶ Powerful labeling technique
 - Reveals enzyme activity, particle / drug location etc.
- ▶ Non-invasive
 - Does not require subject to be euthanized
- ▶ Comply with 3R principle
- ▶ Reduce individual differences
- ▶ Easy Operation
- ▶ Multi-function and Multiplex
 - Bioluminescence, Fluorescence and Cherenkov
- ▶ High Throughput

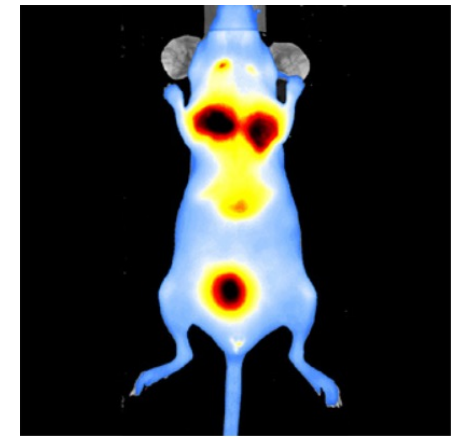
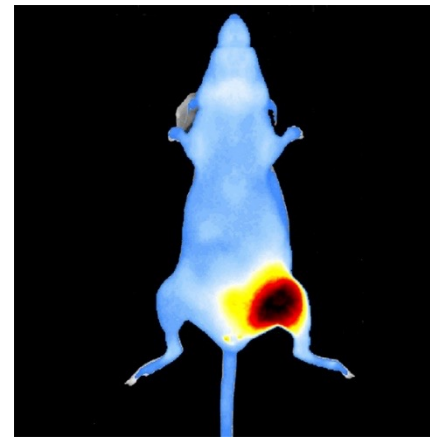
► Bioluminescence & Fluorescent Proteins

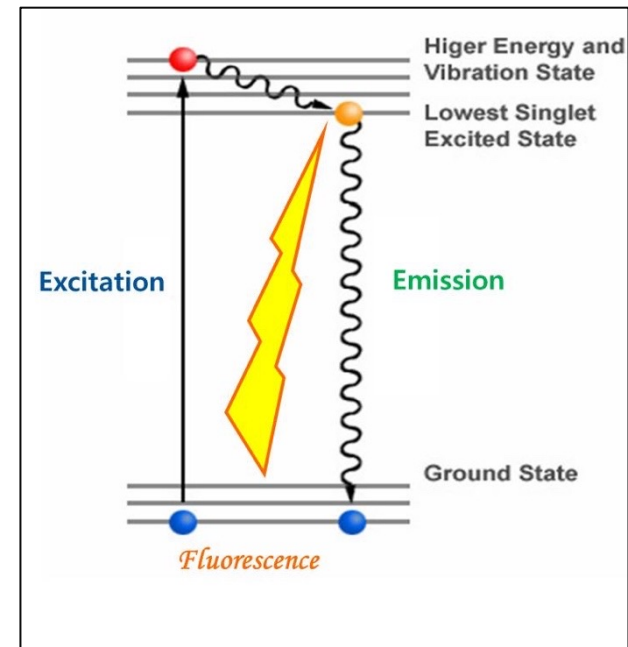
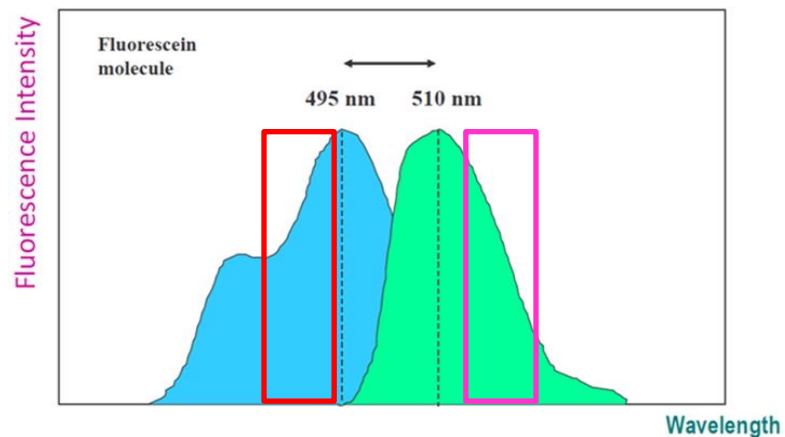
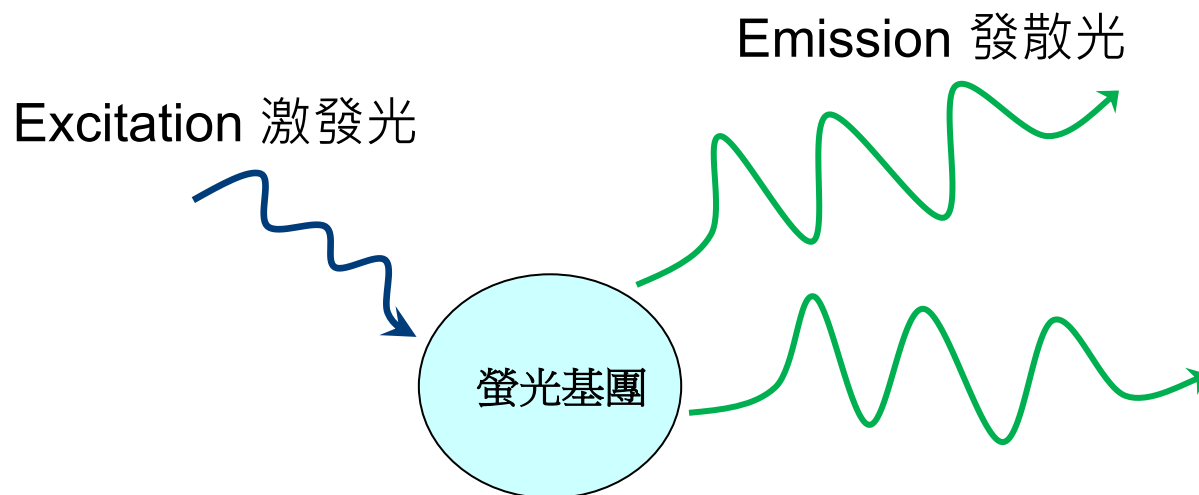
- Powerful approach using animals/cells with modified genetics
- Uses promoter systems for deep understanding of underlying mechanisms

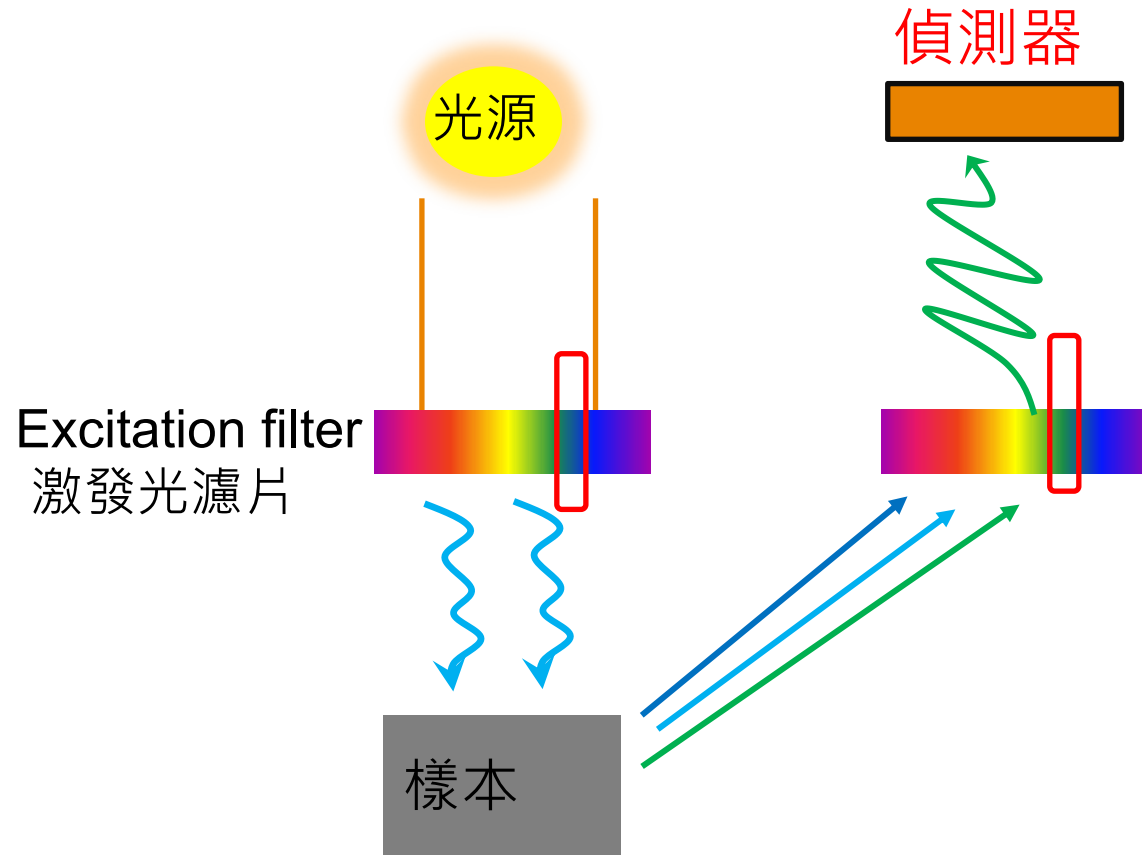


► Fluorescent Agents (Red/NIR)

- Standard disease biology/models
- Injectable drug-like imaging agents to view biology



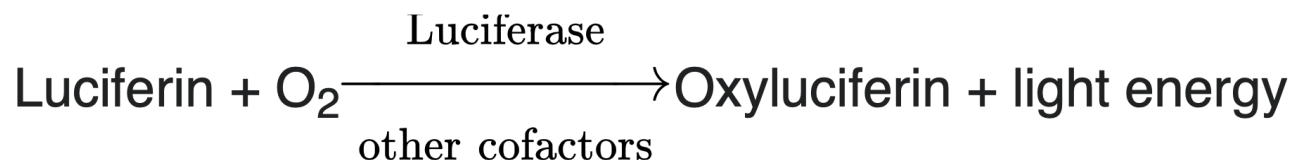




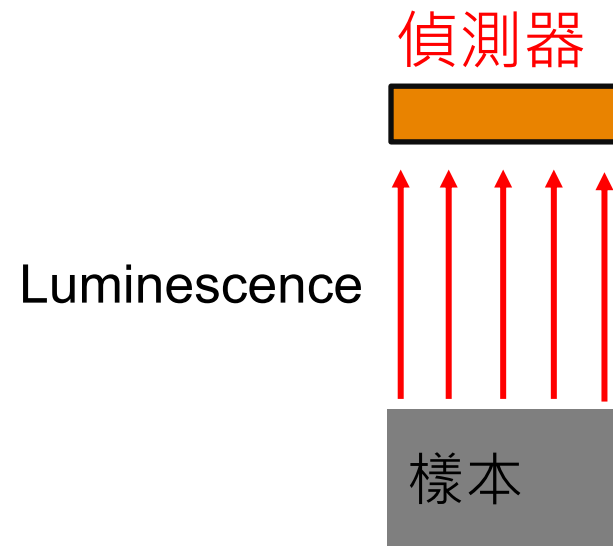
Mice expressing enhanced green fluorescent protein (eGFP) under UV-illumination



In: BMC Cancer. 2012, 12:21. doi:10.1186/1471-2407-12-21 PDF, CC BY 2.0,
<https://commons.wikimedia.org/w/index.php?curid=18345103>

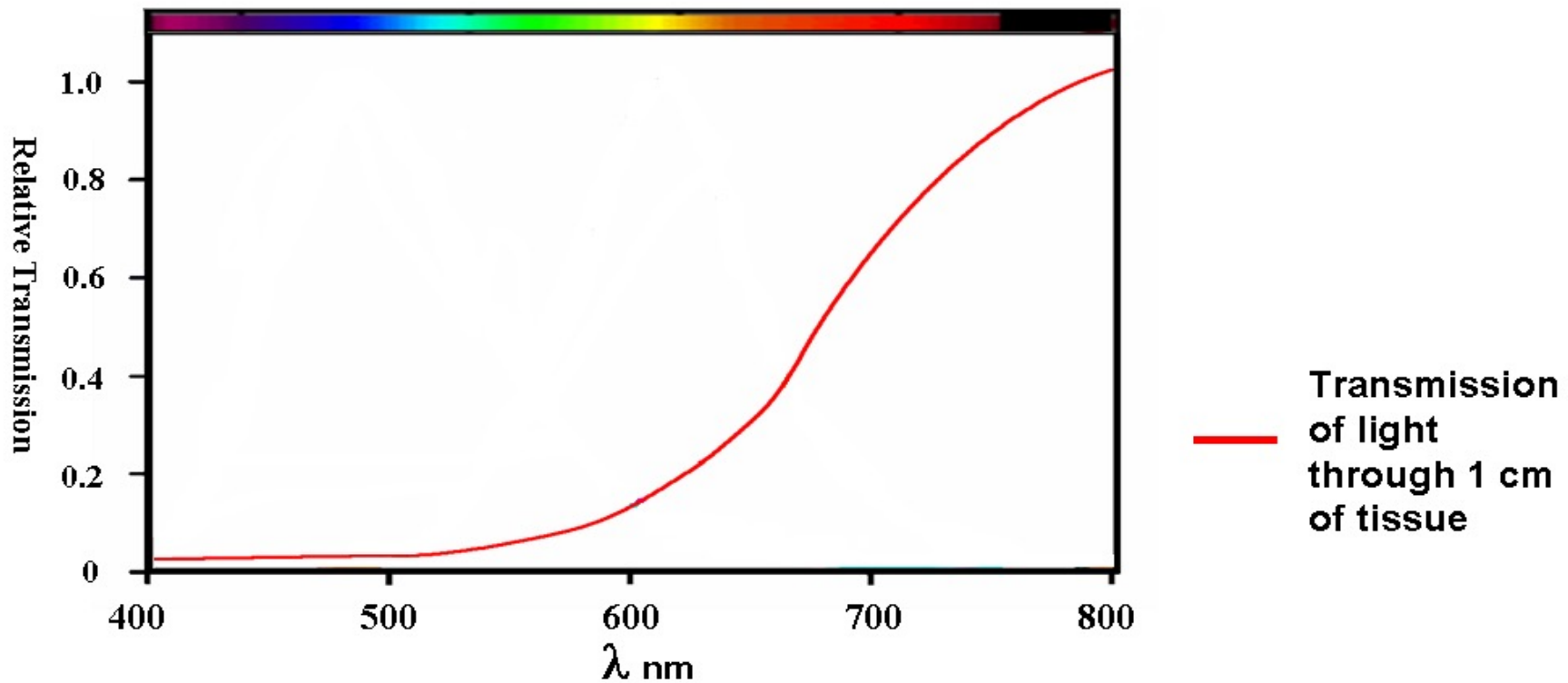


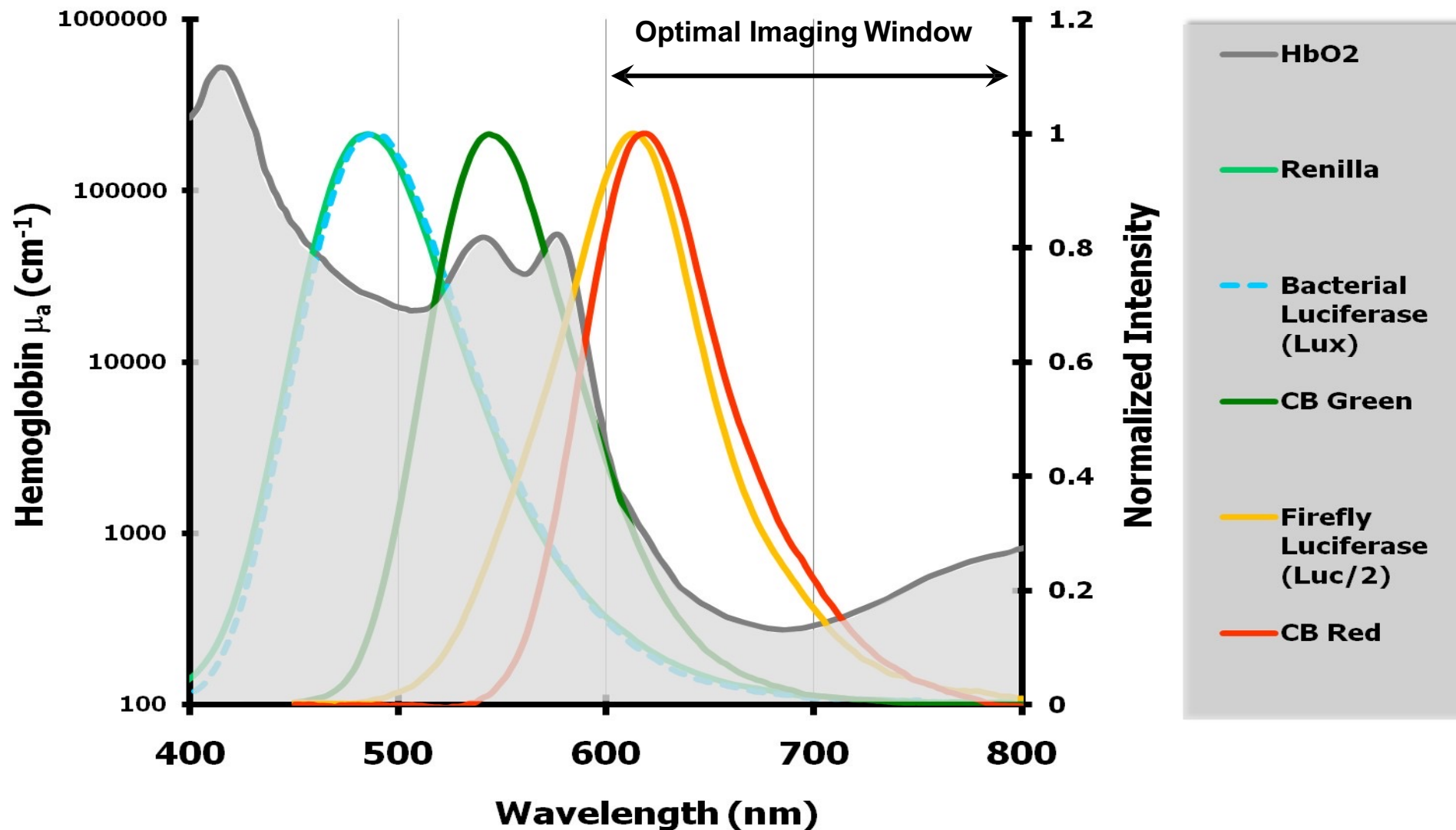
http://media.photobucket.com/image/luminescence%20firefly/Groovy23_Photos/Firefly385_359183a.jpg

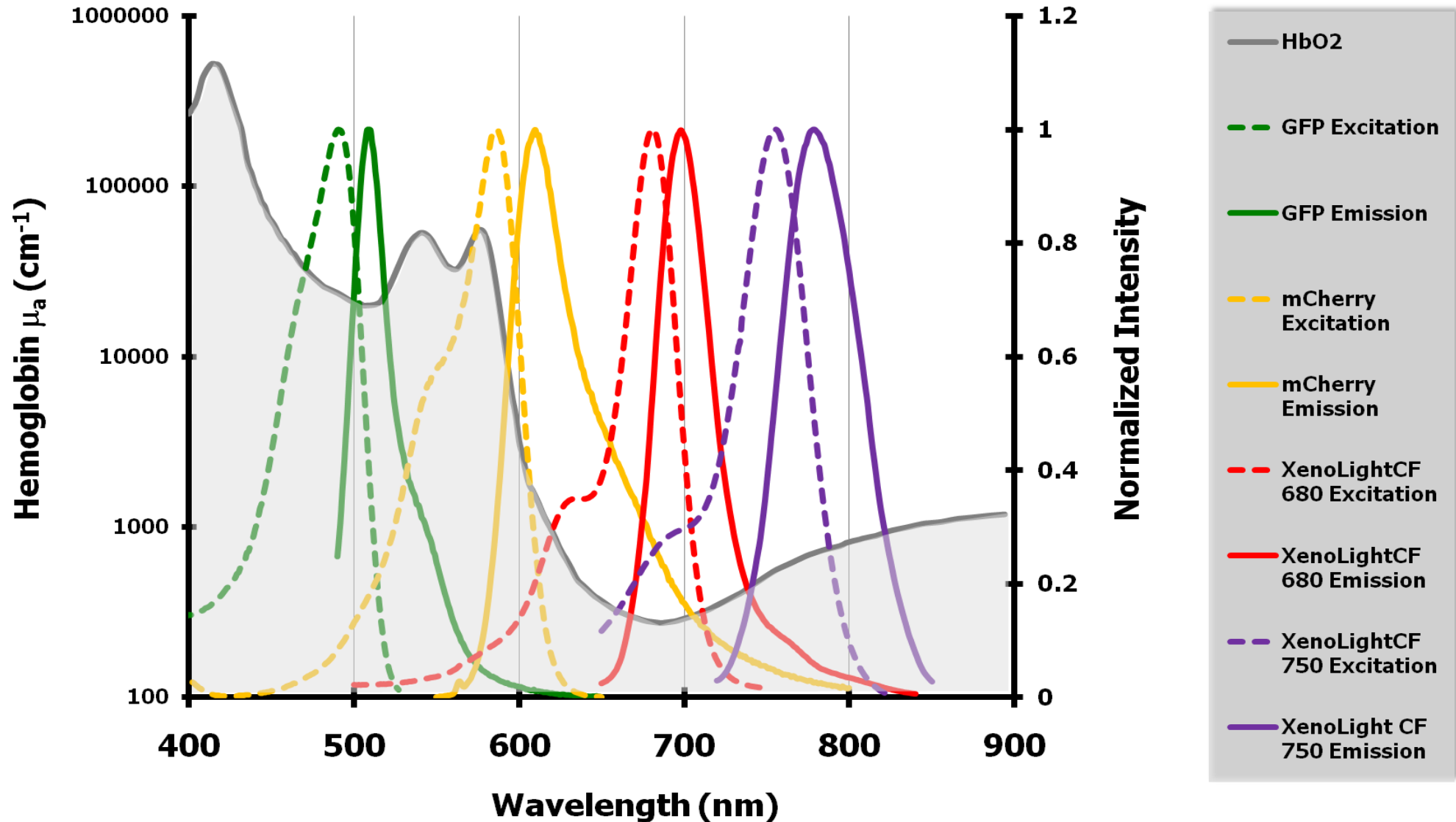


Sample emits light without excitation

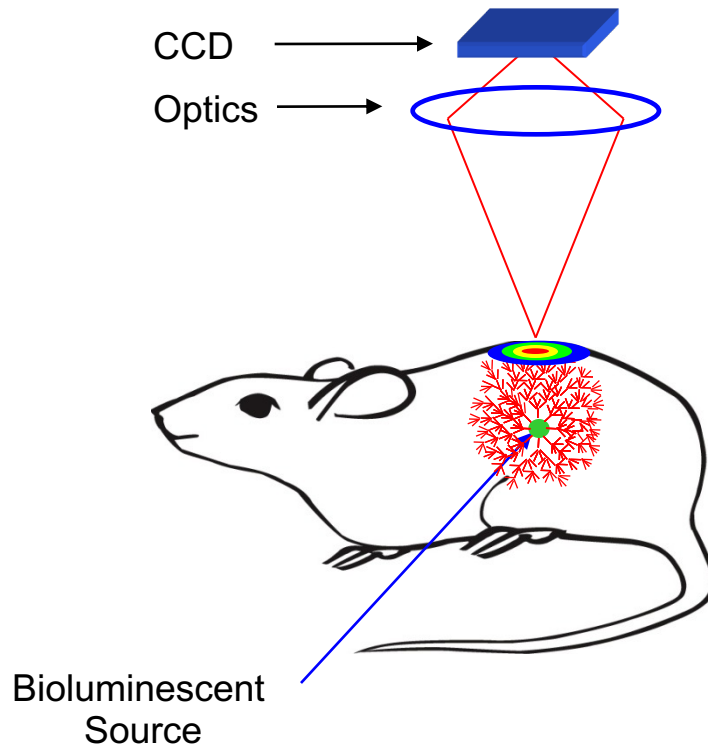
Tissue is not Transparent – Light Absorbance Depends on Wavelength







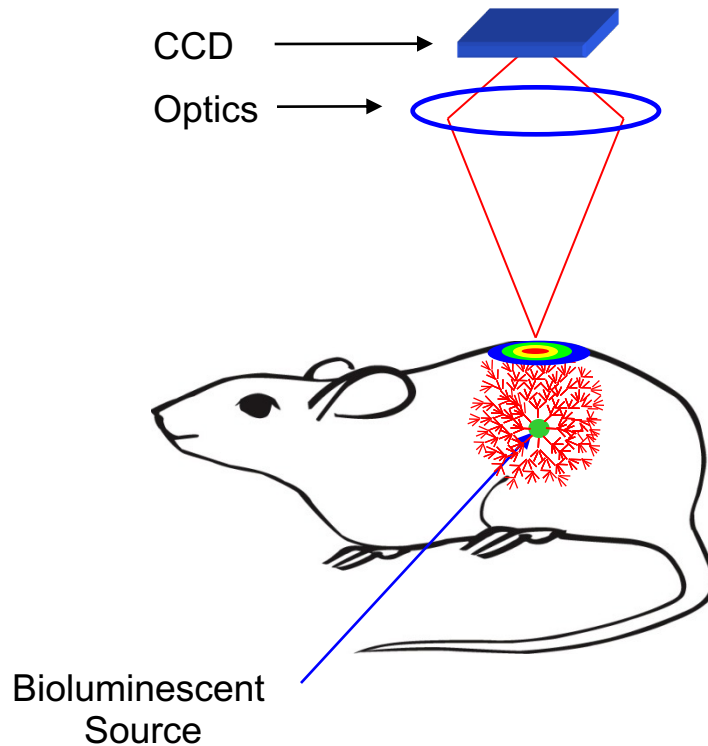
Photons Diffuse Through Tissue and the IVIS[®] Views this Signal on the Surface of the Subject



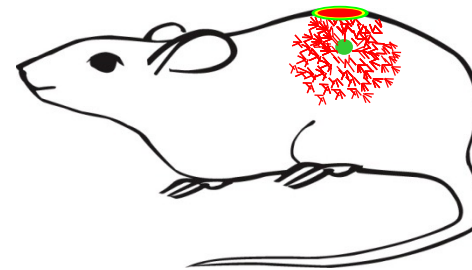
- Light traveling through tissue scatters many times creating a "fuzzy" image at the surface of the animal
- The IVIS[®] views the diffuse image on the surface of the subject

Tissue Diffusion

Photons Diffuse Through Tissue and the IVIS[®] Views this Signal on the Surface of the Subject

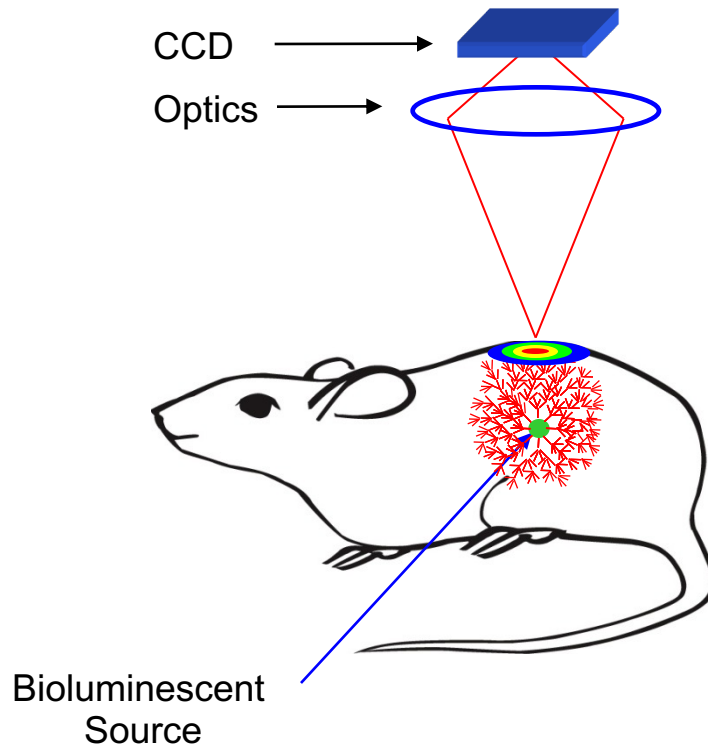


- Light traveling through tissue scatters many times creating a "fuzzy" image at the surface of the animal
- The IVIS[®] views the diffuse image on the surface of the subject



Tissue Diffusion

Photons Diffuse Through Tissue and the IVIS[®] Views this Signal on the Surface of the Subject



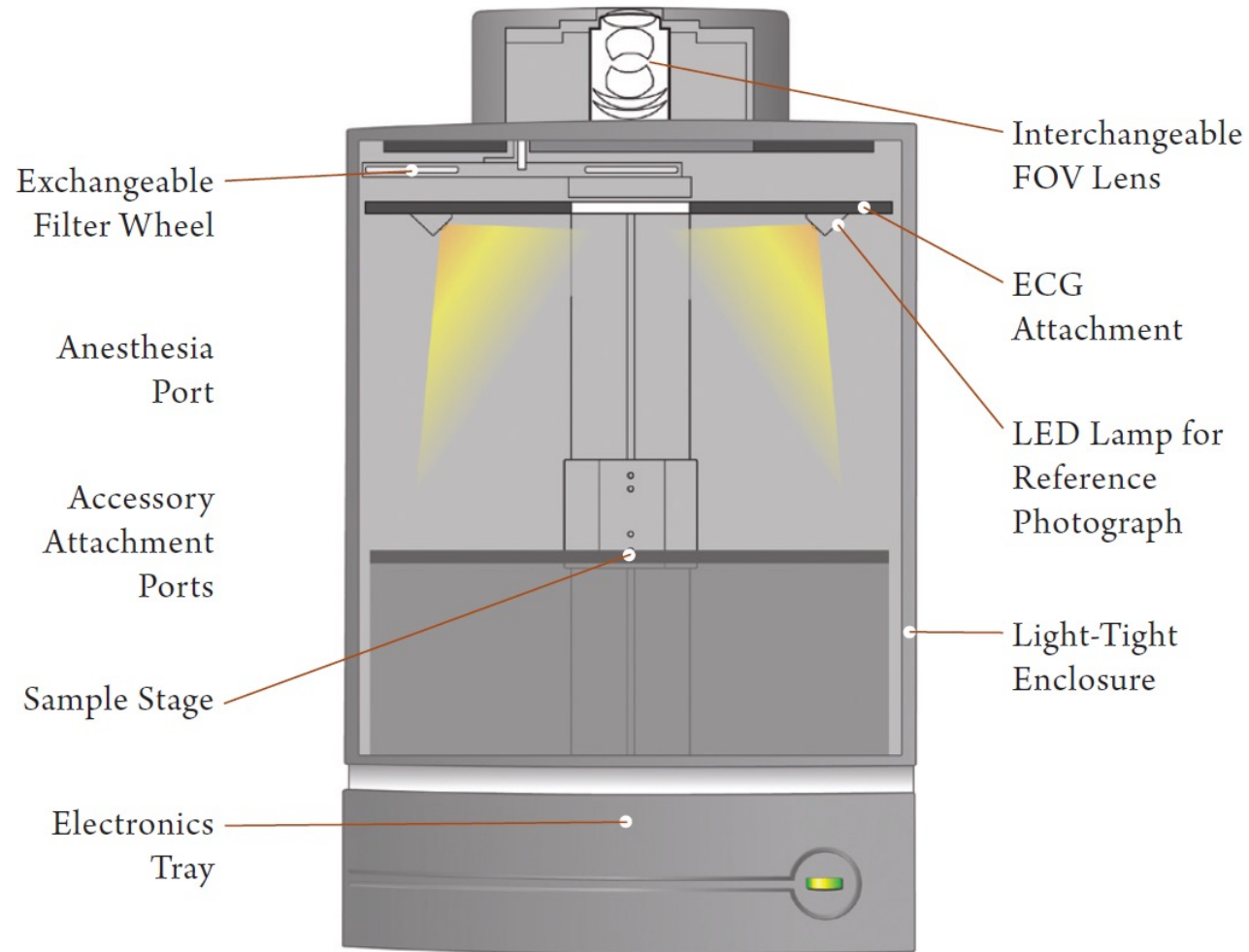
- Light traveling through tissue scatters many times creating a "fuzzy" image at the surface of the animal
- The IVIS[®] views the diffuse image on the surface of the subject





- Customized for *in-vivo* imaging
- High sensitivity detection from 500 - 840 nm
- Large dynamic range

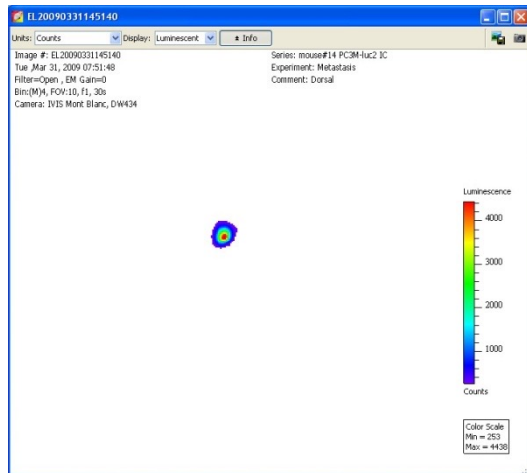
The IVIS[®] Imaging Chamber



- Controls all settings in the IVIS[®] system (fully computer controlled)
- Provides advanced cataloging and browsing tools
- Provides analysis tools for quantification
- Instrument settings are analogous to photography
- Images are acquired in a two step process

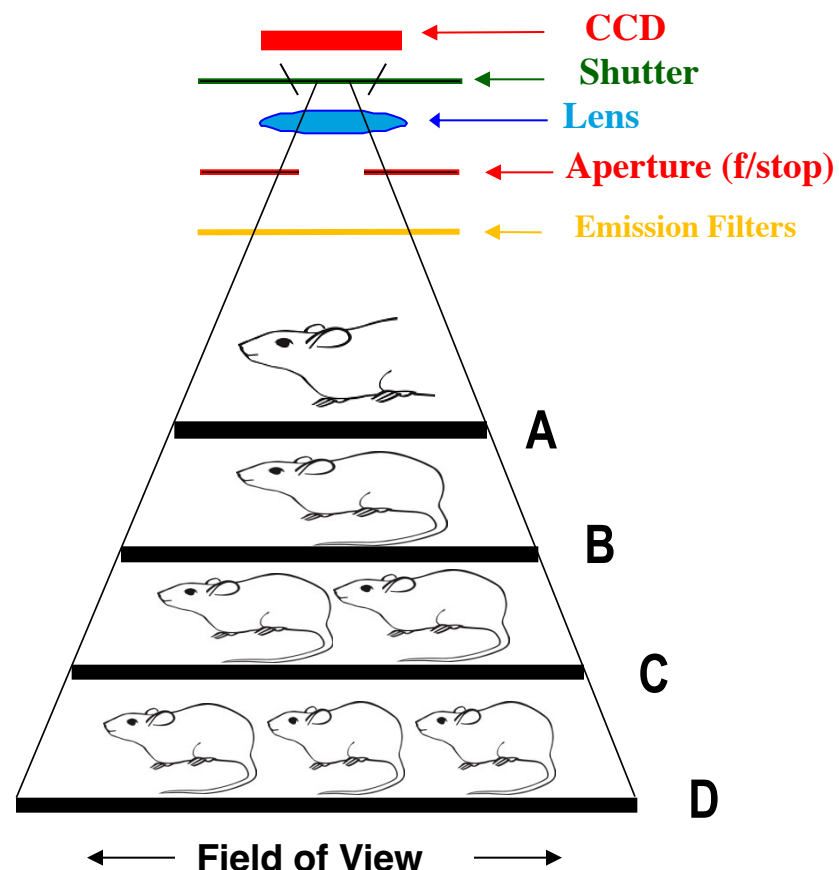
Standard Acquisition Overlay

Composed of Two Images
Photographic + Luminescent = Overlay



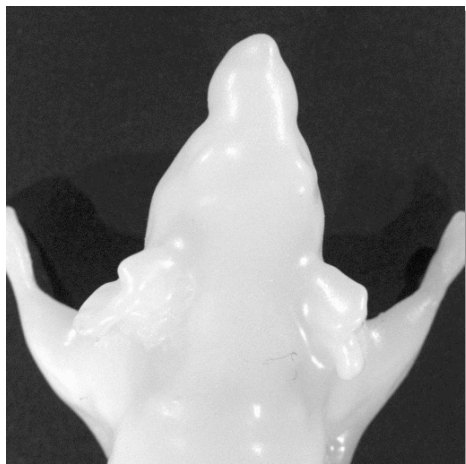
Camera and Lens Settings are Analogous to Those Used in Standard Photography

- Field of View (FOV) is dependent on the distance from the lens to the sample
- Light collected is proportional to how long the shutter is open (exposure time)
- Aperture (f/stop) controls the amount of light collected
- Digital pixel binning possible with CCD - for further increase in sensitivity



Field-of-View (typical)

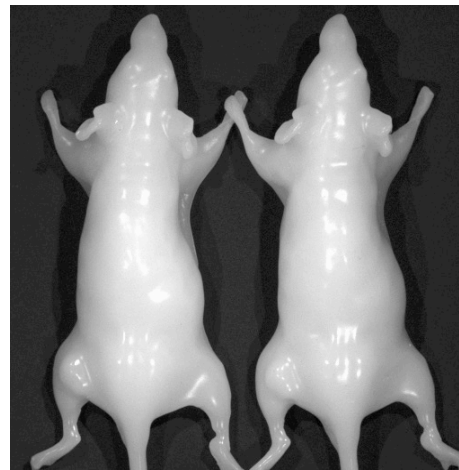
FOV A = 5 x 5 cm



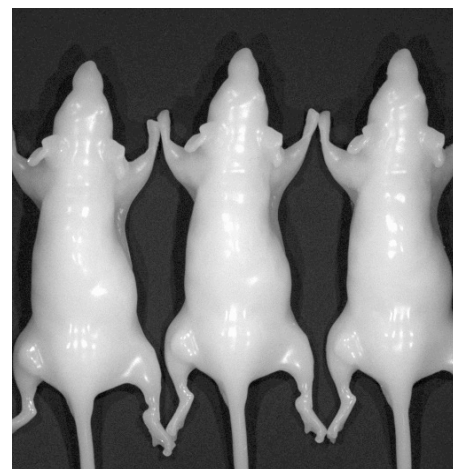
FOV B = 7.5 x 7.5 cm



FOV C = 10 x 10 cm



FOV D = 12.5 x 12.5 cm

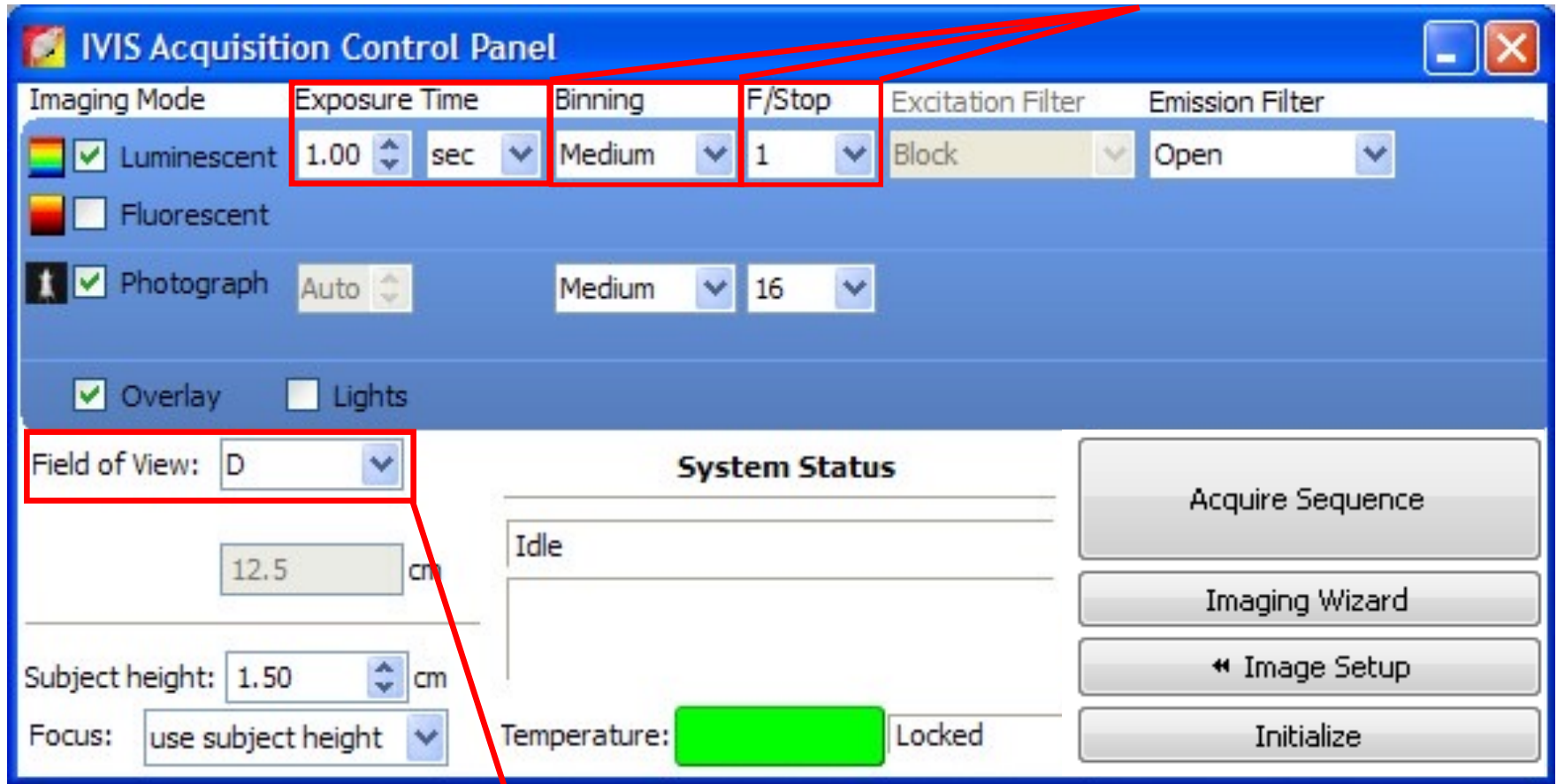


Setting Sensitivity – Luminescent Signal Level

- The IVIS® CCD camera has a raw signal range of 0 to 65535 Analog to Digital Counts (2^{16}).
- Adjust camera settings to obtain a signal level of 600 to 60,000 counts.
- Settings that control signal level are:
 - Exposure time
 - Binning (CCD Resolution)
 - f/stop (Aperture)
- Instrument is calibrated to automatically compensate for changes in sensitivity settings

Living Image Control Panel

Controls Sensitivity



The screenshot shows the IVIS Acquisition Control Panel. A red box highlights the 'Exposure Time' (1.00 sec), 'Binning' (Medium), and 'F/Stop' (1) settings. Another red box highlights the 'Field of View' (D) setting. A red arrow points from the 'Field of View' box to the 'Temperature' field, which is highlighted in green. The 'System Status' section shows 'Idle' and 'Locked'.

IVIS Acquisition Control Panel

Imaging Mode: ☒ Luminescent ☐ Fluorescent ☒ Photograph

Exposure Time: 1.00 sec Binning: Medium F/Stop: 1

Excitation Filter: Block Emission Filter: Open

Auto: Medium 16

☒ Overlay ☐ Lights

Field of View: D

12.5 cm

Subject height: 1.50 cm

Focus: use subject height

System Status

Idle

Temperature: Locked

Acquire Sequence

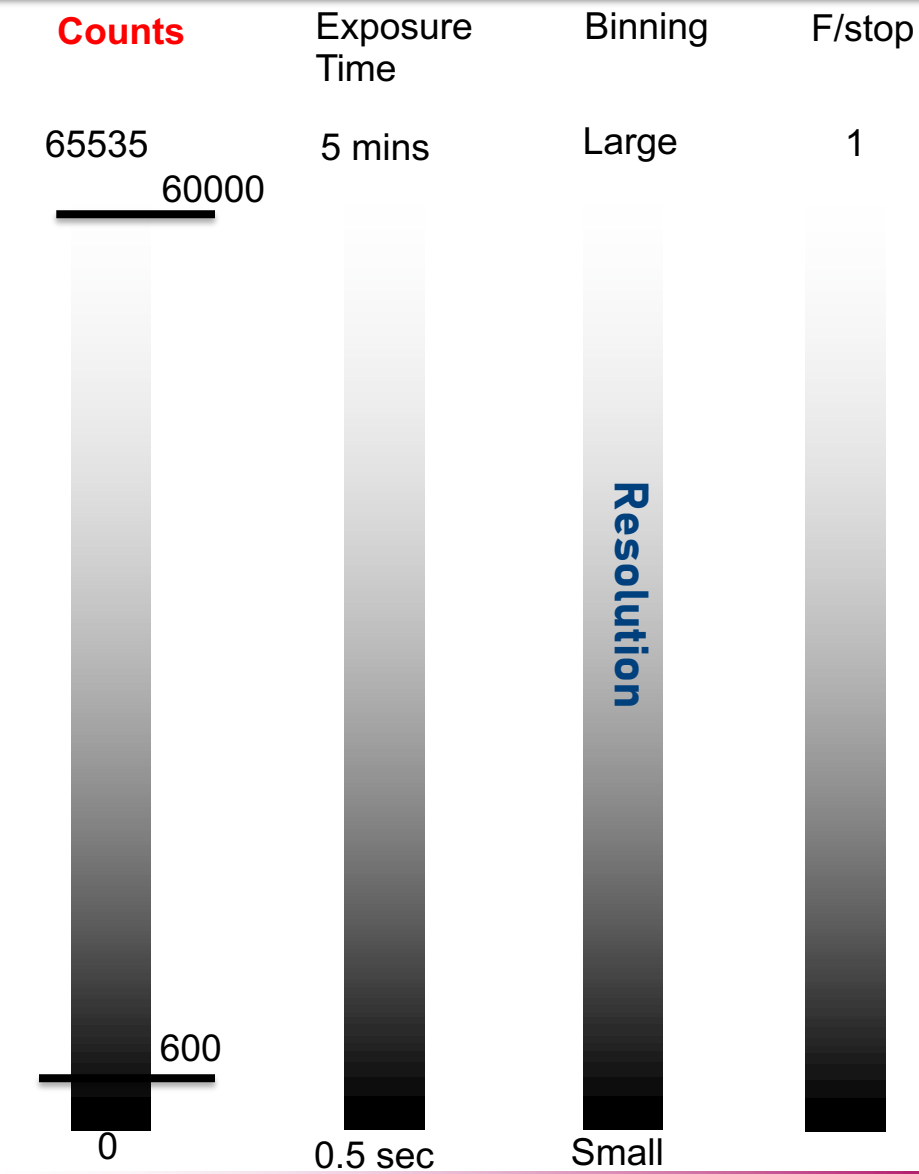
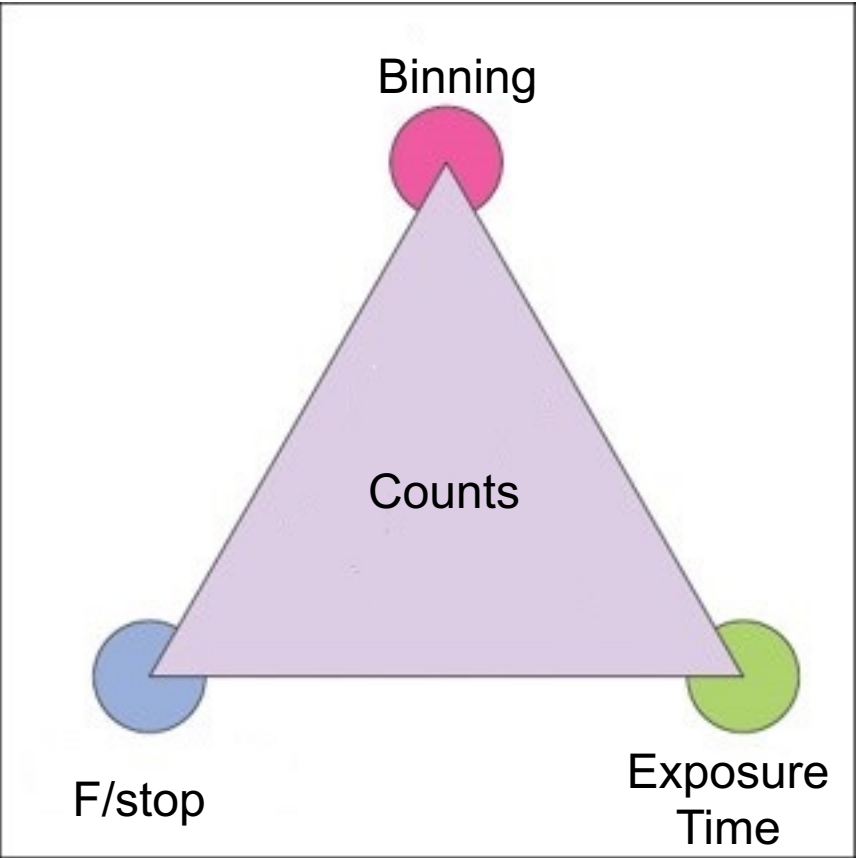
Imaging Wizard

Image Setup

Initialize

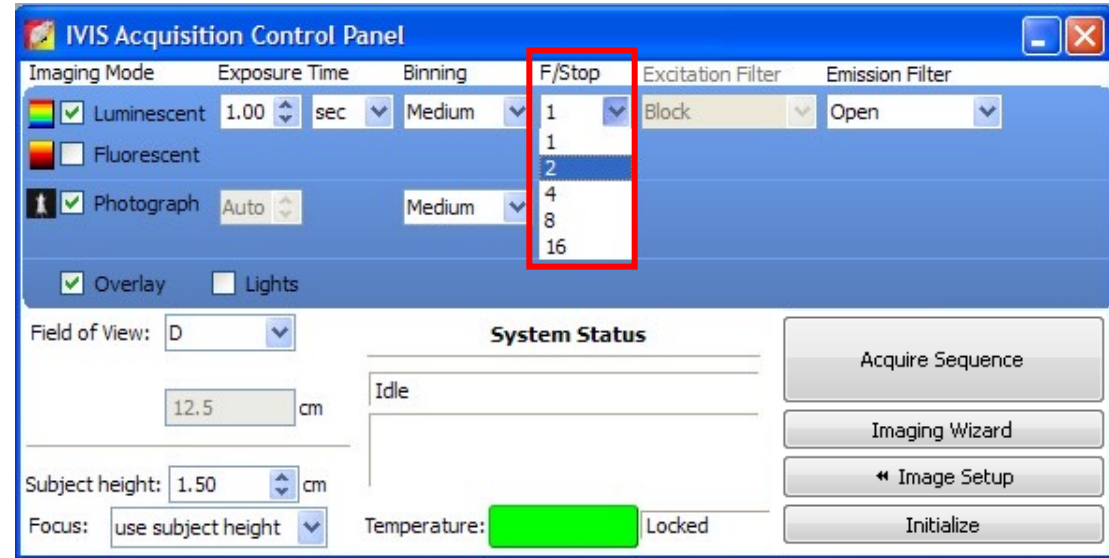
Affects Sensitivity

Sensitivity Controls – Relationship of F/Stop, Exposure Time and Bining

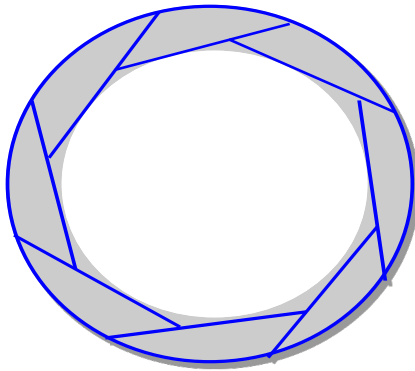


f/stop (Lens Aperture)

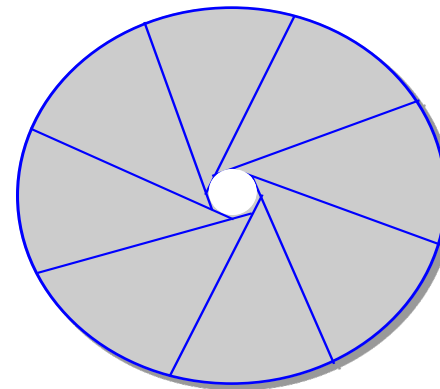
- f/stop controls the amount of light received by the CCD
- f/1 is wide open, maximum light collection - default for luminescent
- f/16 is smallest aperture, best resolution - default for photo



f/1

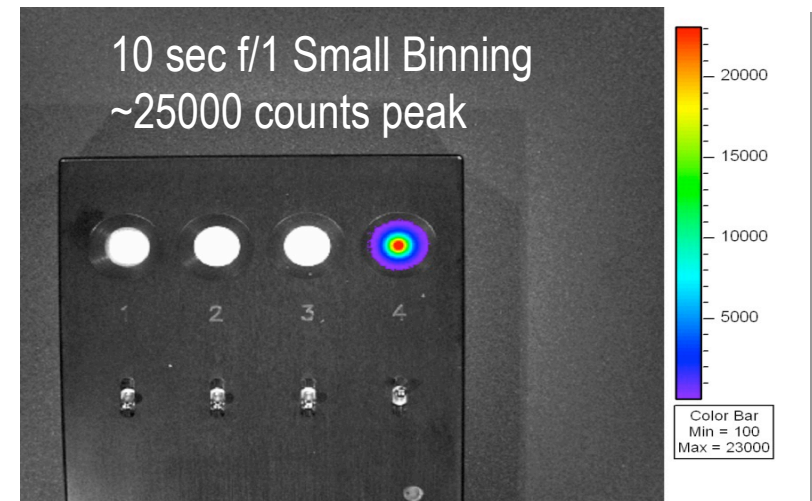
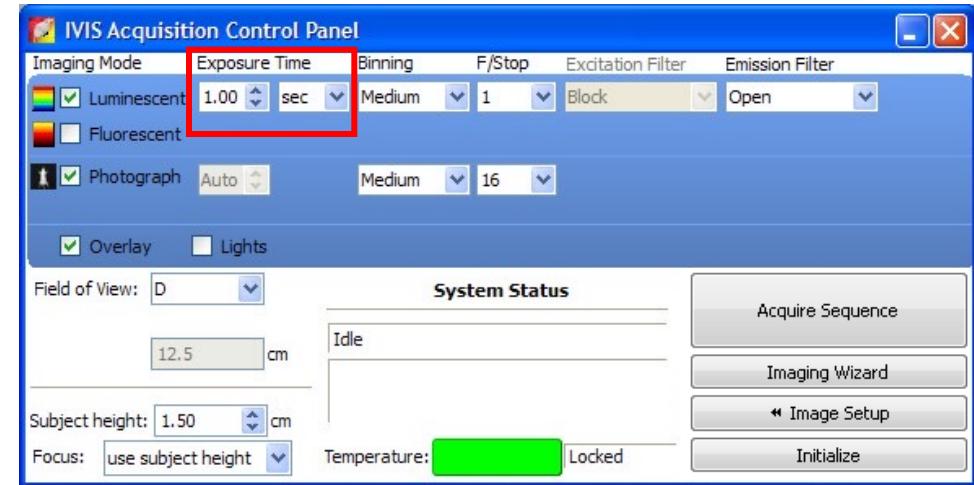


f/16



Exposure Time

- Signal level is directly proportional to exposure time
- Shorter exposure time improves throughput
(Recommended min exposure time > 0.5 secs)
- Longer exposure time increases signal
(Recommended max exposure time < 5 mins)



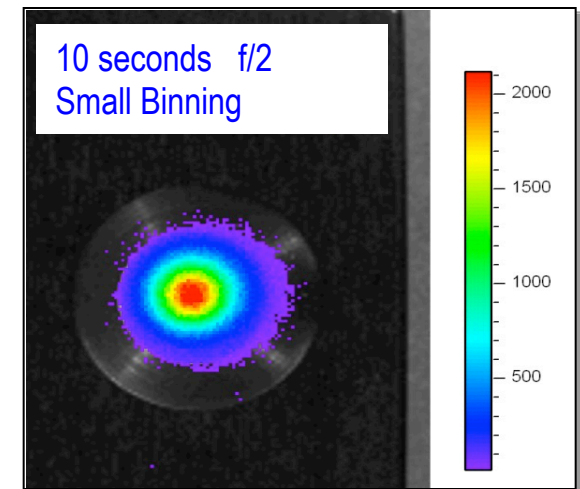
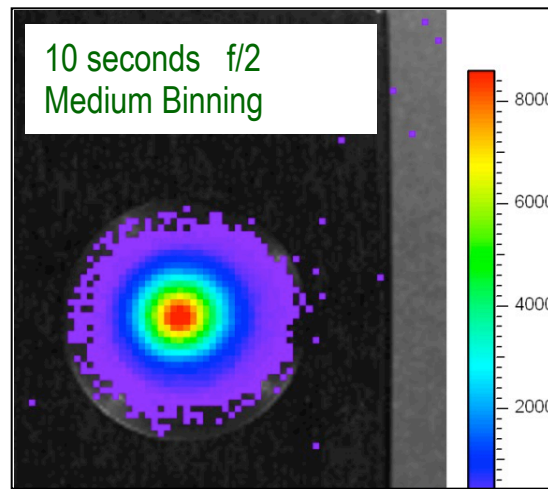
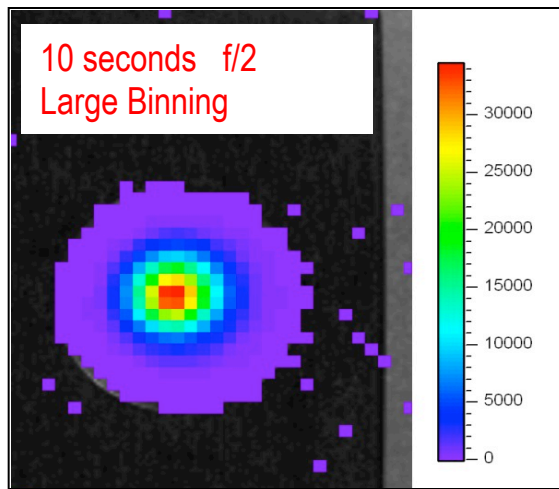
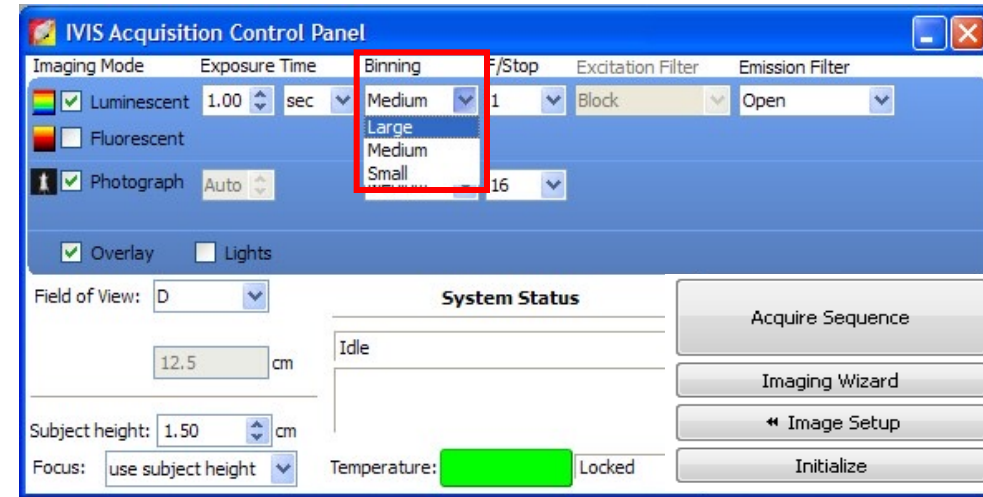
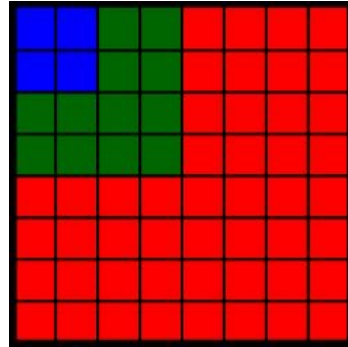
Pixel Binning (CCD Resolution)

- Binning refers to the grouping of pixels into a larger super-pixel

Large Binning (8)
Higher Sensitivity/
Lower Resolution

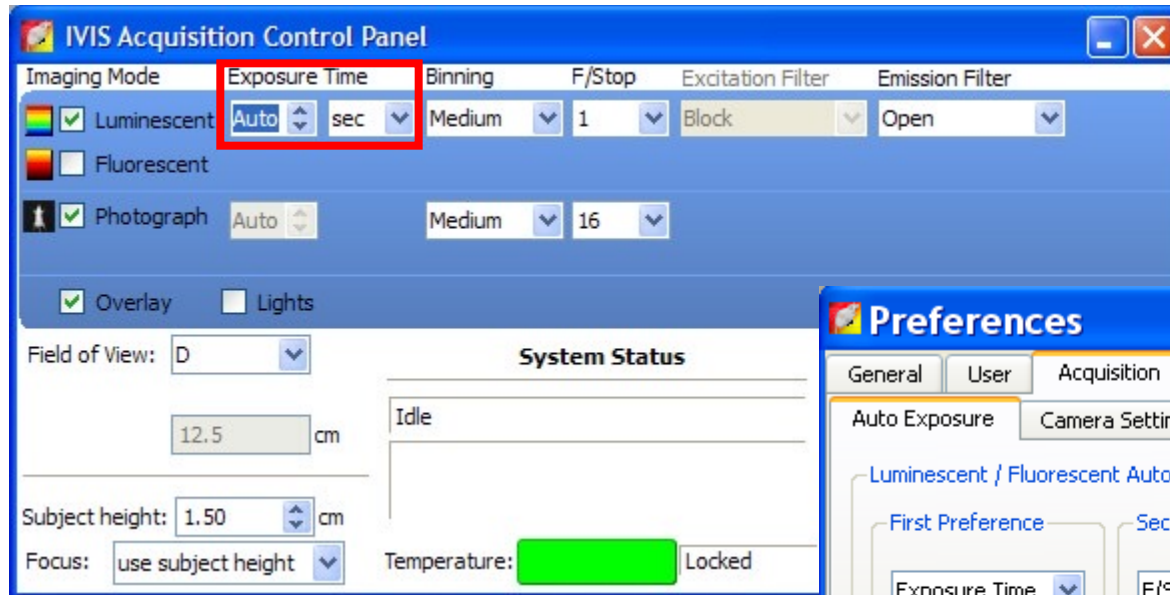
Medium Binning (4)

Small Binning (2)
Higher Resolution /
Lower Sensitivity

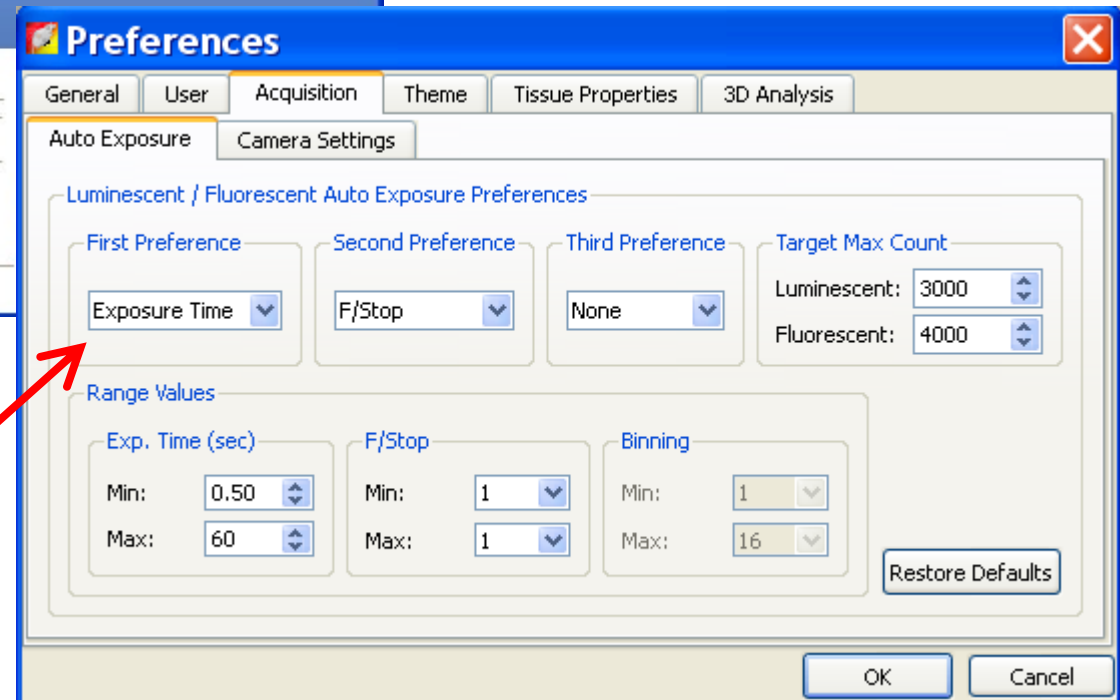


Auto-Exposure

Auto exposure feature available for bioluminescence and fluorescence

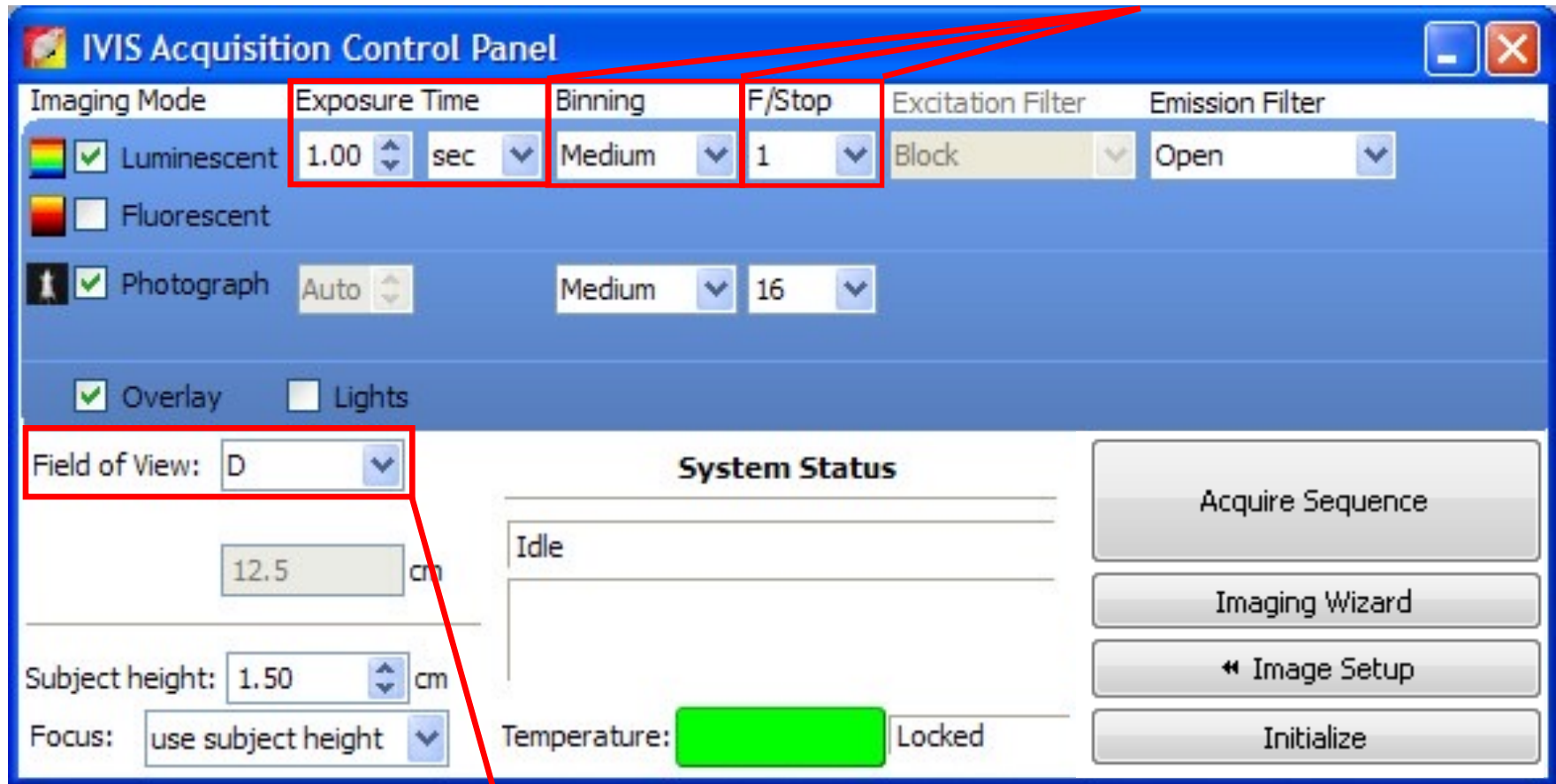


User definable auto-exposure settings



Summary of Basic Camera Settings

Controls Sensitivity



The screenshot shows the IVIS Acquisition Control Panel with several settings highlighted by red boxes and arrows. The 'Exposure Time' is set to 1.00 sec, 'Binning' is set to Medium, and 'F/Stop' is set to 1. The 'Field of View' is set to D, which is linked to a distance of 12.5 cm. The 'Subject height' is set to 1.50 cm, and the 'Focus' is set to 'use subject height'. The 'System Status' shows 'Idle' and 'Temperature' is 'Locked'. The 'Acquire Sequence' button is highlighted in green.

Imaging Mode	Exposure Time	Binning	F/Stop	Excitation Filter	Emission Filter
☒ Luminescent	1.00 sec	Medium	1	Block	Open
☐ Fluorescent					

Photograph: ☒ Auto Medium 16

Overlay: ☒ Lights: ☐

Field of View: D 12.5 cm

Subject height: 1.50 cm

Focus: use subject height

System Status: Idle

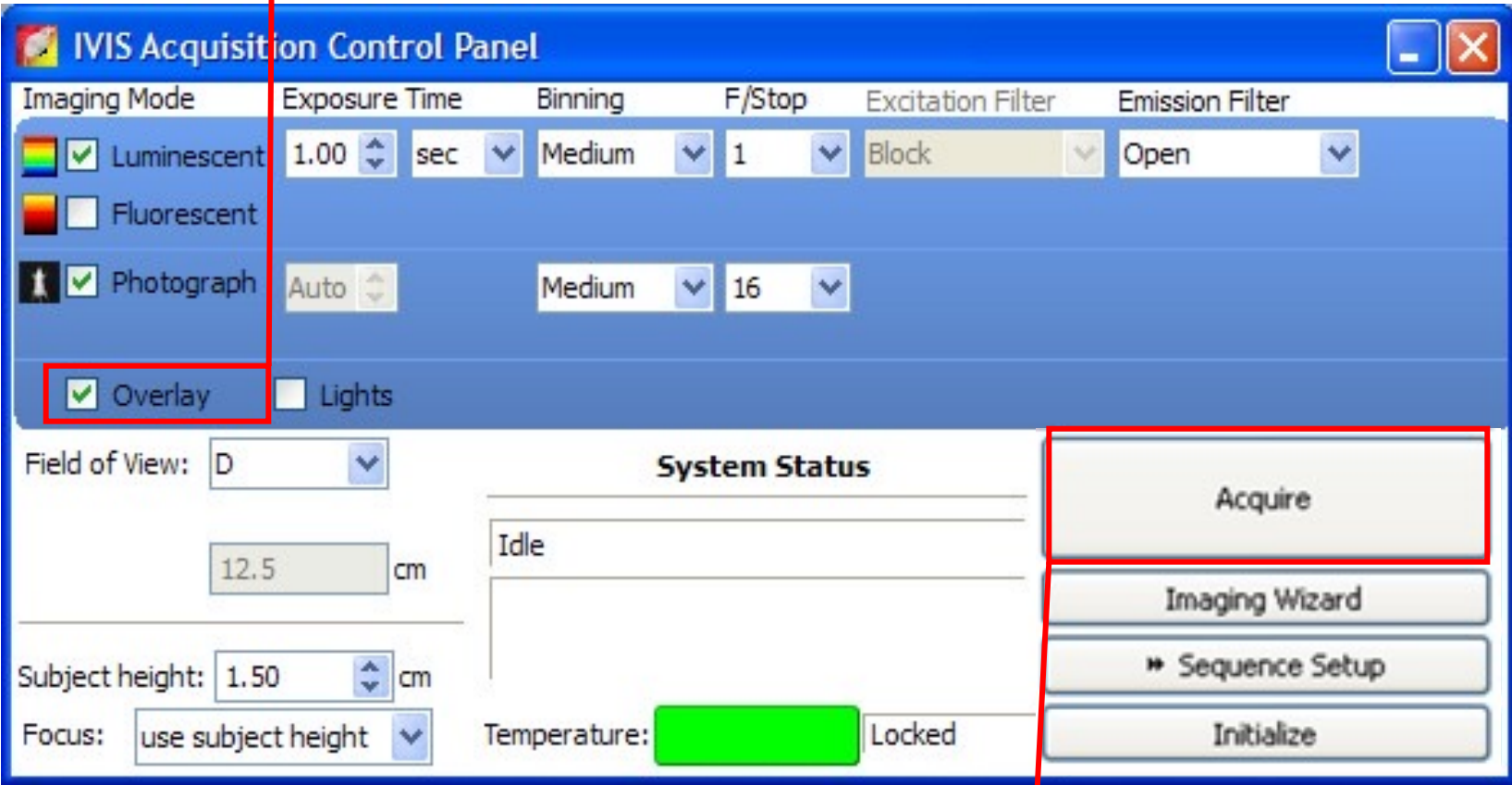
Temperature: Locked

Buttons: Acquire Sequence, Imaging Wizard, Image Setup, Initialize

Affects Sensitivity

Acquisition – Single Image

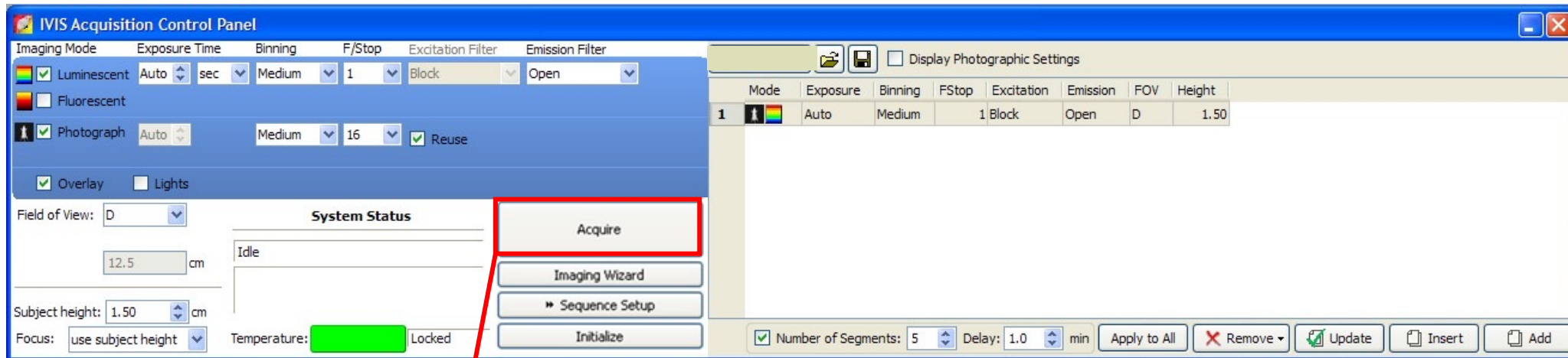
Overlay will automatically take Photo + Luminescent



Single Image Acquisition

Acquisition – Sequential mode

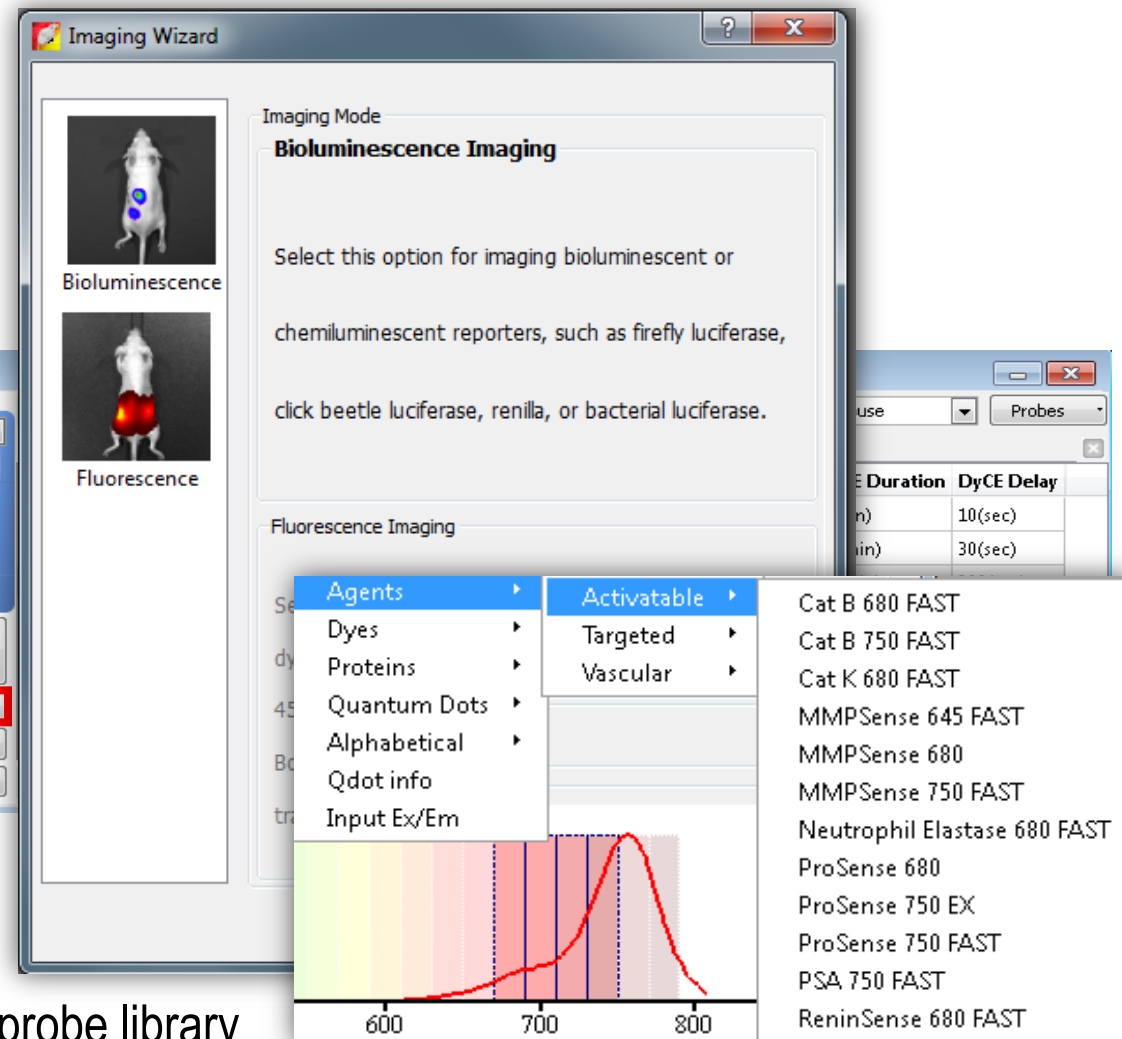
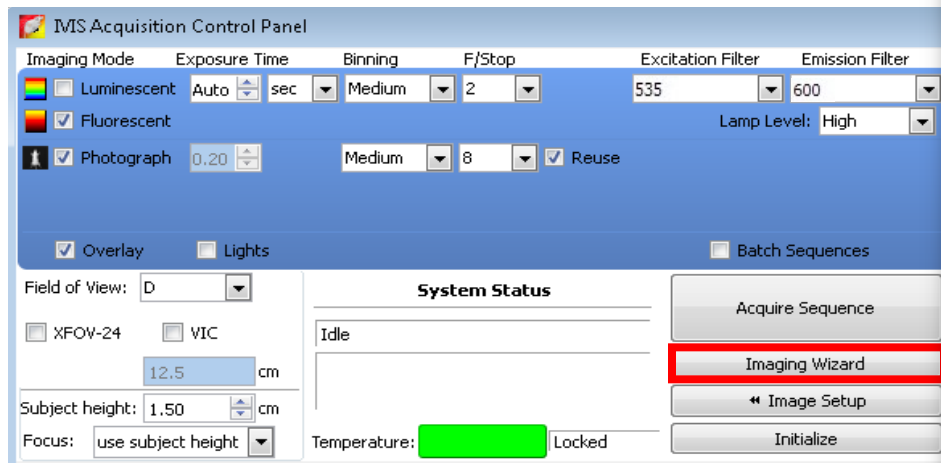
Allows automatic acquisition of a series of images separated by fixed time points.



Starts Sequential
Image Acquisition

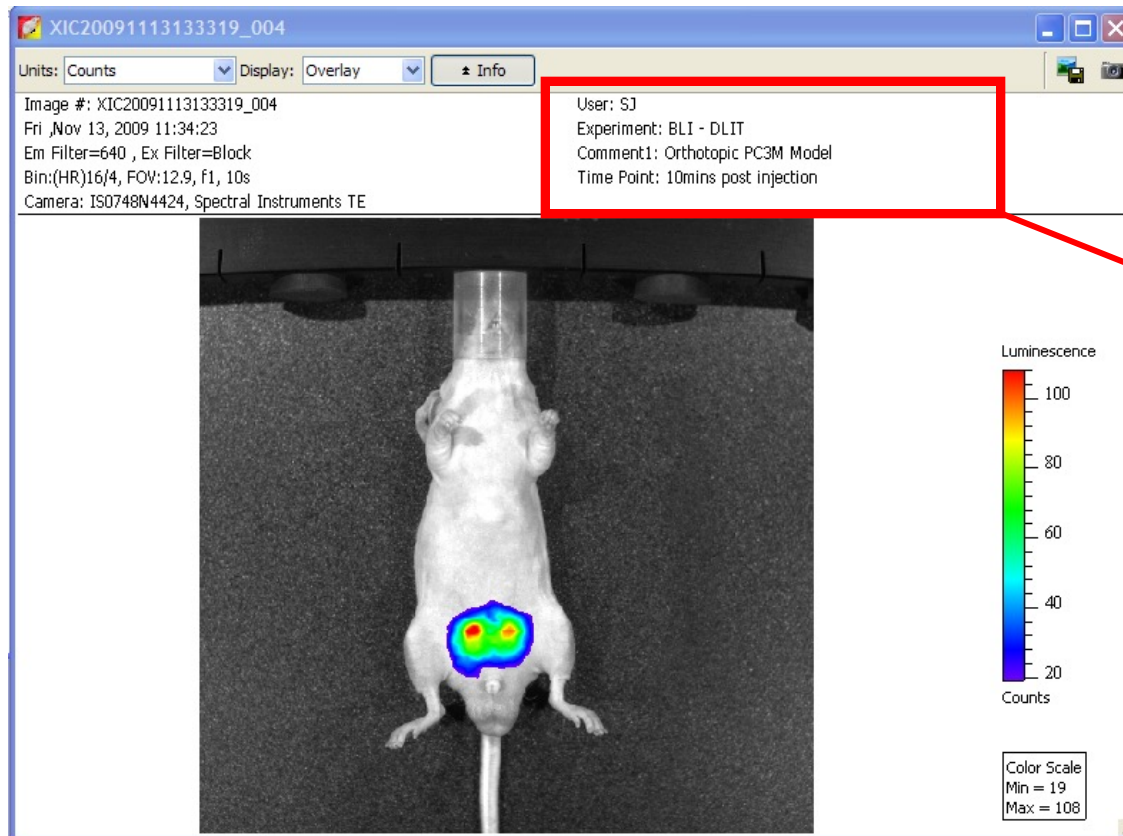
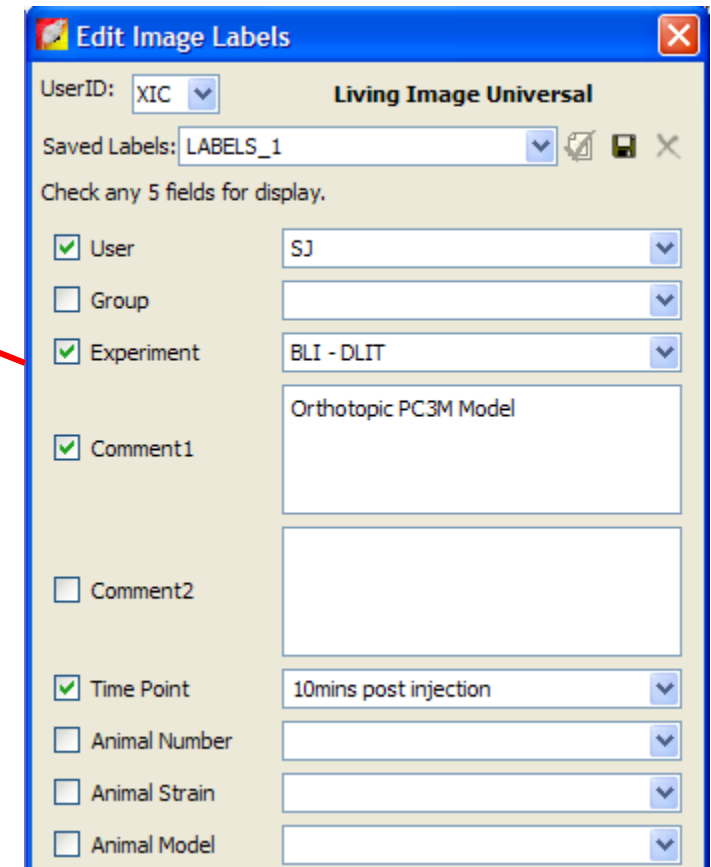
User Friendly
Sequence Editor

Select for assistance in setting up bioluminescence or fluorescence sequences



Expanded probe library

Good labeling practices are necessary for effective data browsing

Edit Image Labels

UserID: XIC Living Image Universal

Saved Labels: LABELS_1

Check any 5 fields for display.

☒ User SJ

☐ Group

☒ Experiment BLI - DLIT
 Orthotopic PC3M Model

☒ Comment1

☐ Comment2

☒ Time Point 10mins post injection

☐ Animal Number

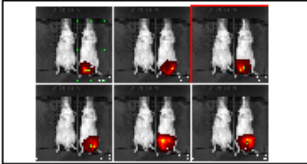
☐ Animal Strain

☐ Animal Model

Image Browsing

Living Image® Browser

XIC20091113121810_SEQ



Click Number	Date and Time	Experiment	Comment1	Comment2	Series	Experiment*
+ SEQ CK20080731152638_SEQ	07/31/2008 15:26:53	1.9e13/uL, 2:1 dilution, 2uL each	XF750 mixed 7/15/08			
+ SEQ CK20080731155429_SEQ	07/31/2008 15:54:56	9.5e12/uL, 2uL each	XF750 mixed 7/15/08	SN GOLD src A		
+ SEQ CK20090330114556_SEQ	03/30/2009 11:46:28	PC3M-tdT Prostate Ortho Avastin-750 injection	Mouse 3, Day 4	FLIT tdT 570/620 dorsal		
+ SEQ CK20090330120010_SEQ	03/30/2009 12:00:47	PC3M-tdT Prostate Ortho Avastin-750 injection	Mouse 3, Day 4	FLIT AF750 745/800 dorsal		
+ SEQ TLT20050624145507_SEQ	06/24/2005 14:55:59				Male Nn/nu	DOB: 03/21/05
+ SEQ XIC20091113091104_SEQ	11/13/2009 09:11:17	Tomato - pC3m				
+ SEQ XIC20091113091618_SEQ	11/13/2009 09:16:32	Tomato - pC3m				
+ SEQ XIC20091113105259_SEQ	11/13/2009 10:53:13					

Hide Browse View Close Preview Label Set: All Add to List Browse.. View: Default Configure Load as Group Load Remove Close

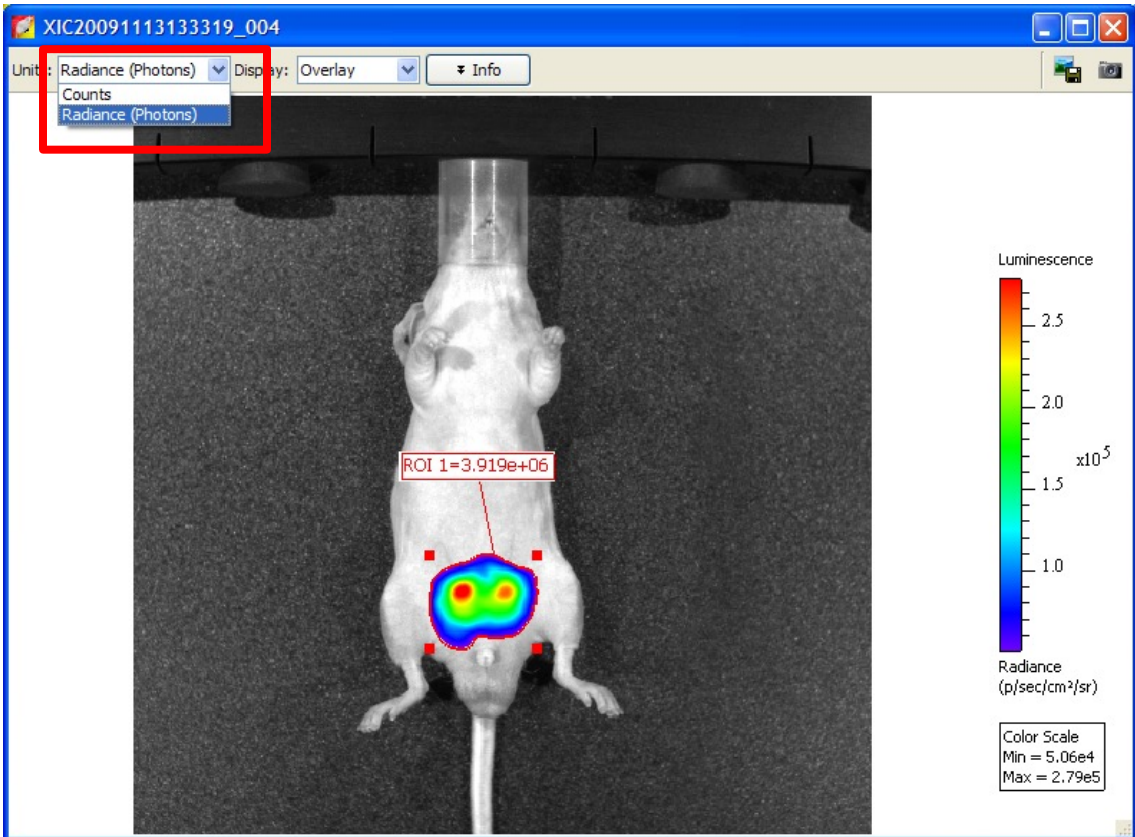
Location: C:/Documents and Settings/joness/Desktop/Xenogen/Training material/LI4 images/SPUM/Pc3m/FLIT/XIC20091113121810_SEQ/SequenceInfo.txt

← User defined information →

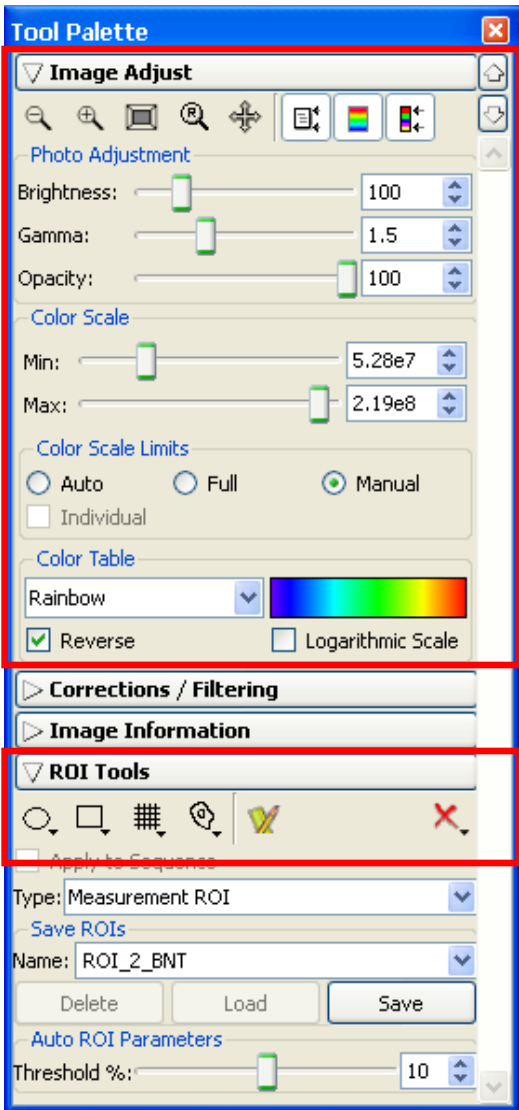
Quantification

- Tool palette for adjusting scale/opacity etc.
- Region of interest (ROI) tools to measure surface intensities

1. 選單位



2. 調整 scale



3. 畫ROI

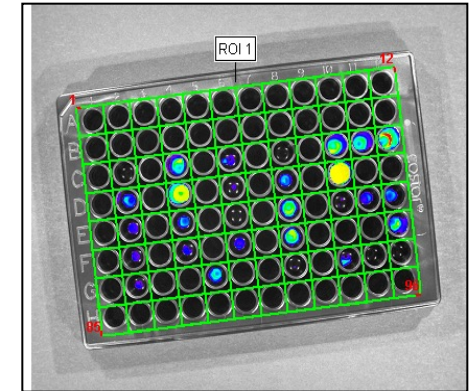
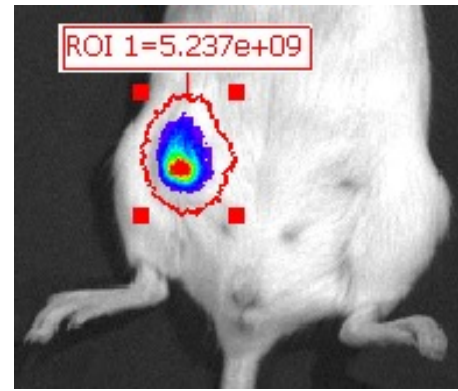
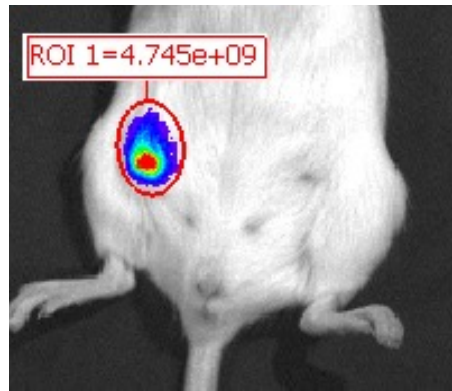
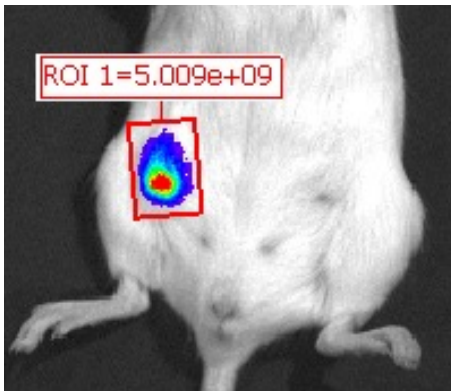
ROI's can be created:

- Manually
- Automatically
- Free Draw

ROI shapes available:

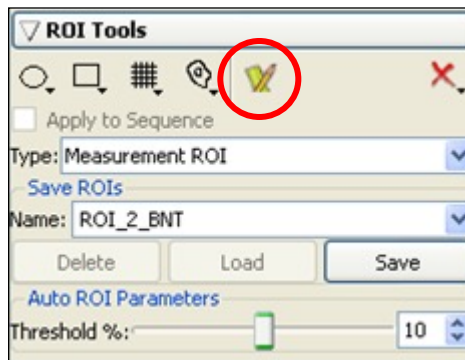
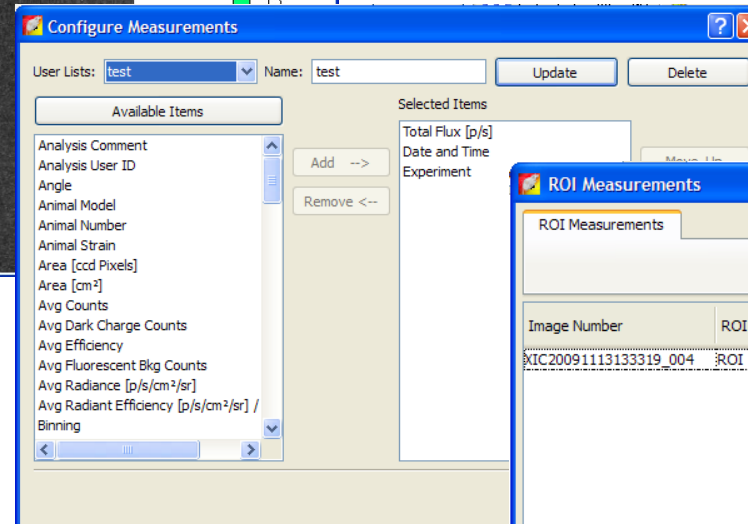
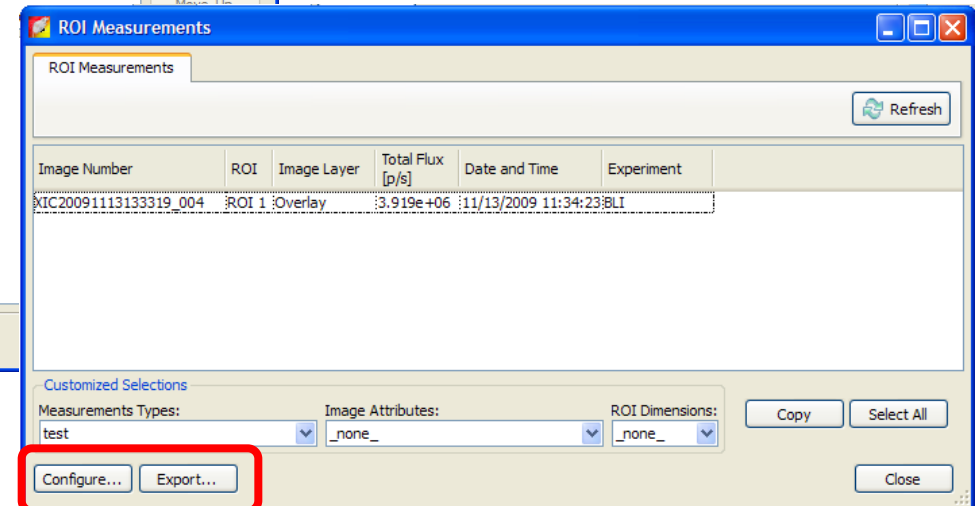
- Square
- Circle
- Contour
- Grid

Important to be consistent with ROI selections



Measurement Table

- Measurement table displays information about each ROI
- Table is user configurable and can be exported to a spreadsheet

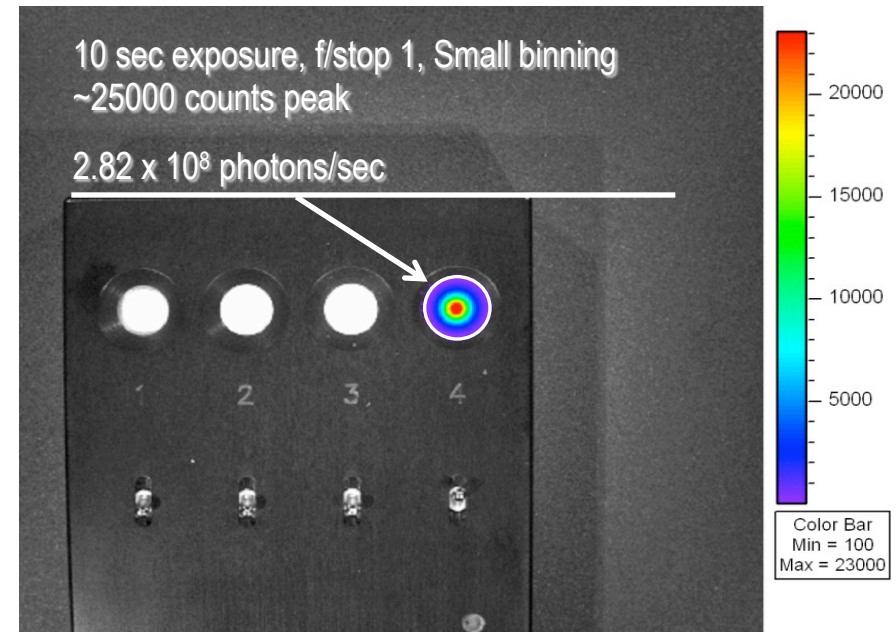
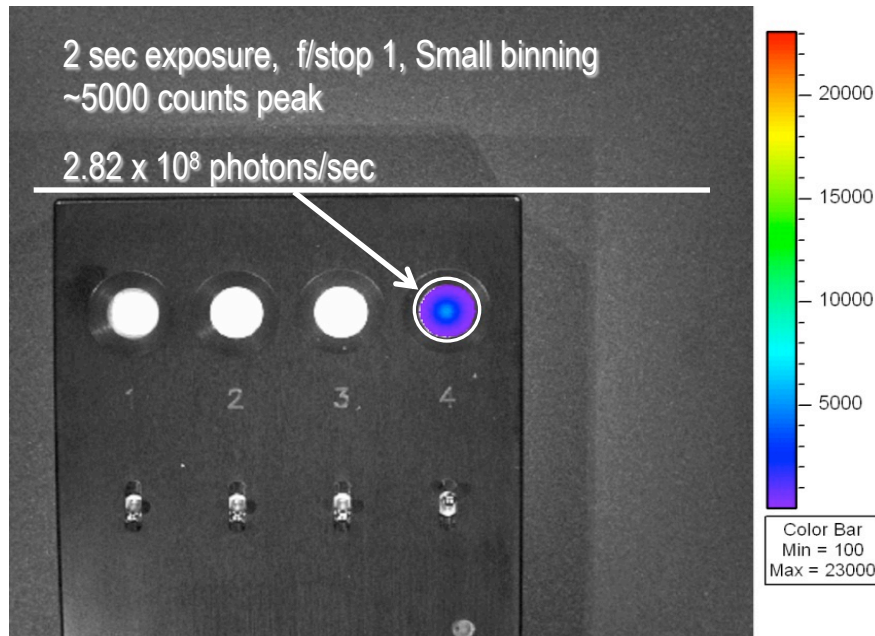
The "ROI Measurements" dialog box displays a table of measurements. The table has columns for "Image Number", "ROI", "Image Layer", "Total Flux [p/s]", "Date and Time", and "Experiment". The data row shows "XIC20091113133319_004", "ROI 1 :Overlay", and "3.919e+06". The "Date and Time" is "11/13/2009 11:34:23BLI".

Image Number	ROI	Image Layer	Total Flux [p/s]	Date and Time	Experiment
XIC20091113133319_004	ROI 1 :Overlay		3.919e+06	11/13/2009 11:34:23BLI	

Below the table, there are sections for "Customized Selections", "Measurements Types" (set to "test"), "Image Attributes" (set to "_none_"), and "ROI Dimensions" (set to "_none_"). The "Configure..." and "Export..." buttons are highlighted with a red box.

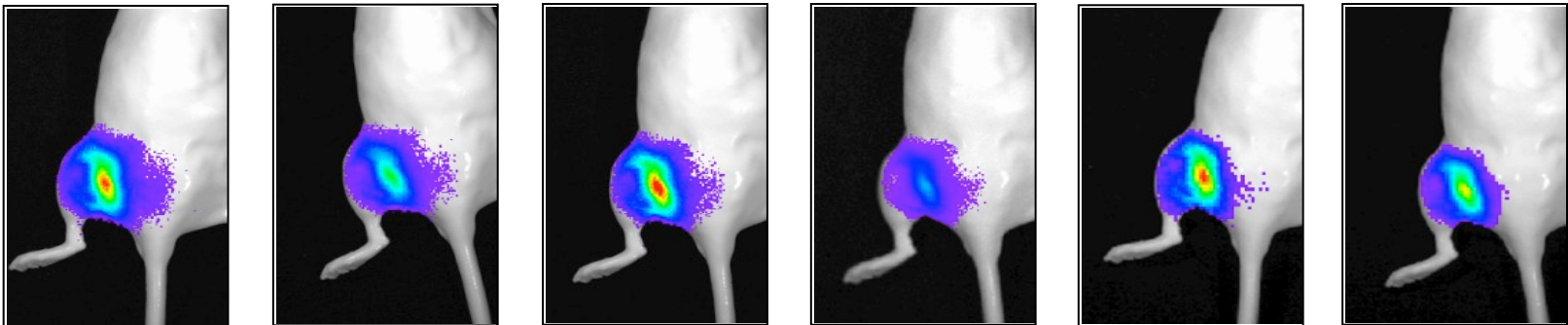
Calibrated Physical Units

- Living Image® automatically compensates for device settings: Exposure time, f/stop, Binning, and Field of View.
- Calibrated units are Photons per Second, representing the flux radiating omni-directionally from a user defined region.



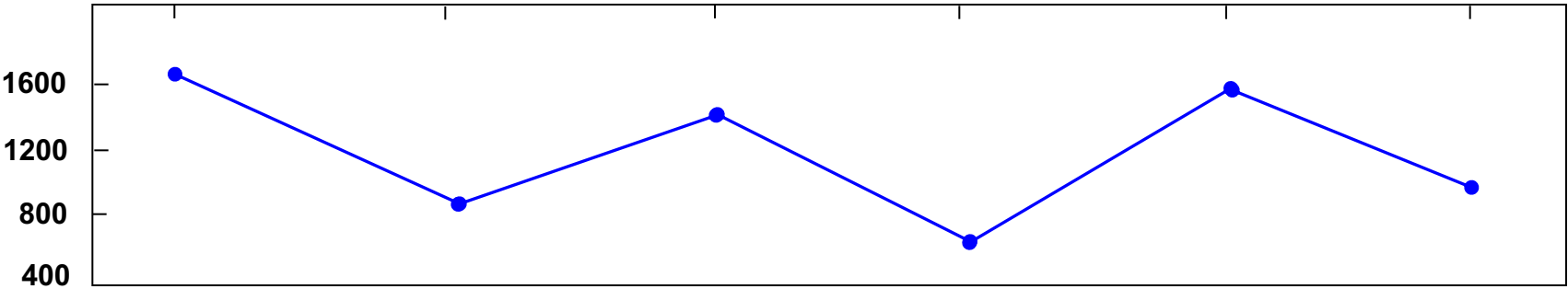
Calibrated Physical Units vs. Raw Signal

Raw Signal
(Counts)



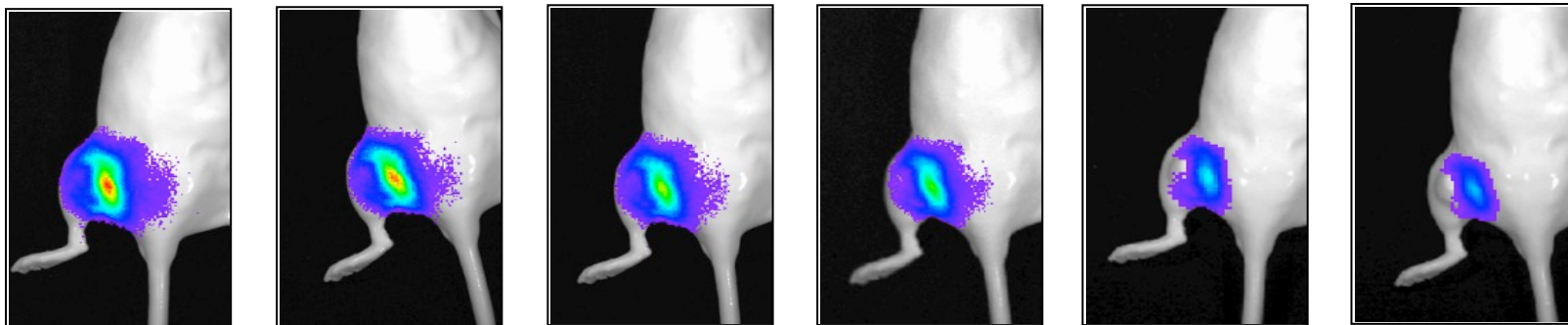
Exp time:	30 sec	30 sec	60 sec	60 sec	60 sec	60 sec
Binning:	small	small	small	small	medium	medium
Day:	1	2	3	4	5	6

Peak
Counts



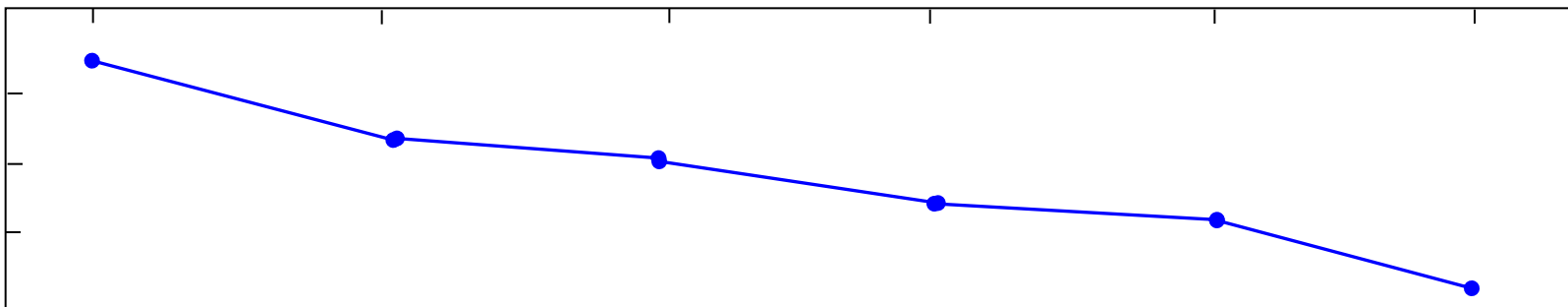
Calibrated Physical Units vs. Raw Signal

**Calibrated
Signal**
*(Photons per
second)*



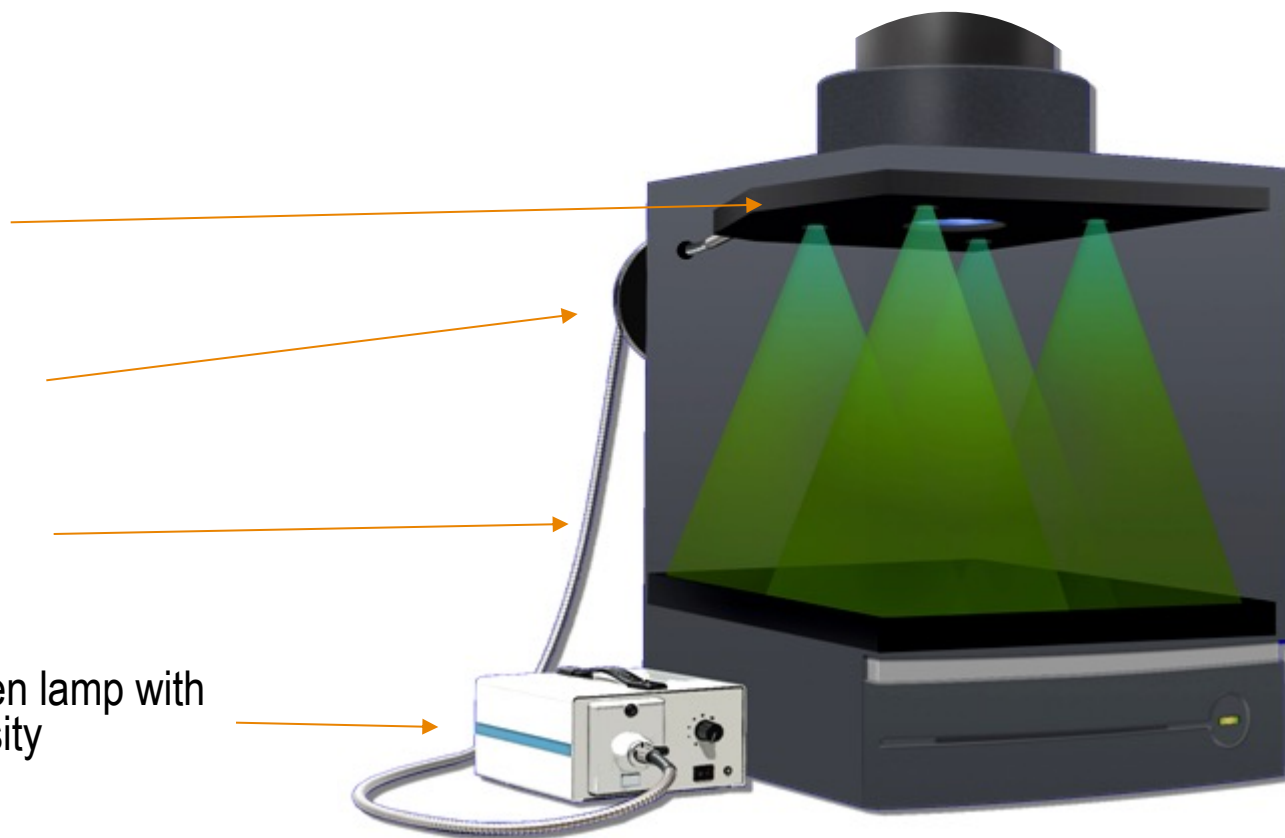
Exp time:	30 sec	30 sec	60 sec	60 sec	60 sec	60 sec
Binning:	small	small	small	small	medium	medium
Day:	1	2	3	4	5	6

Radiance:
*Photons per
second*

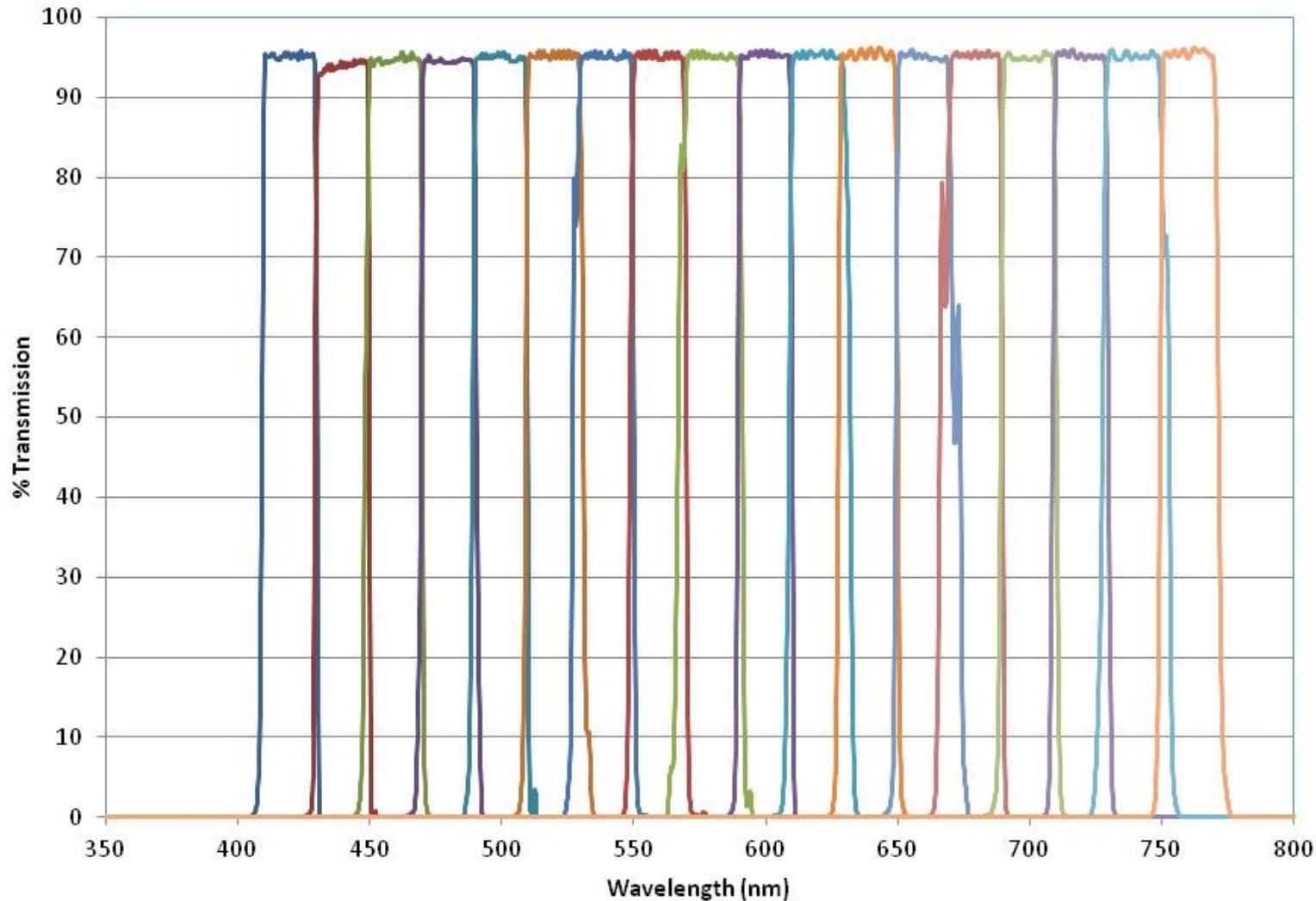


IVIS[®] Lumina's Fluorescence Components

- Fully computer controlled
- Seven position emission filter wheel
- Nineteen position Excitation filter wheel
- Low Autofluorescence optics and fibers
- 150 Watt Tungsten/Halogen lamp with computer controlled intensity

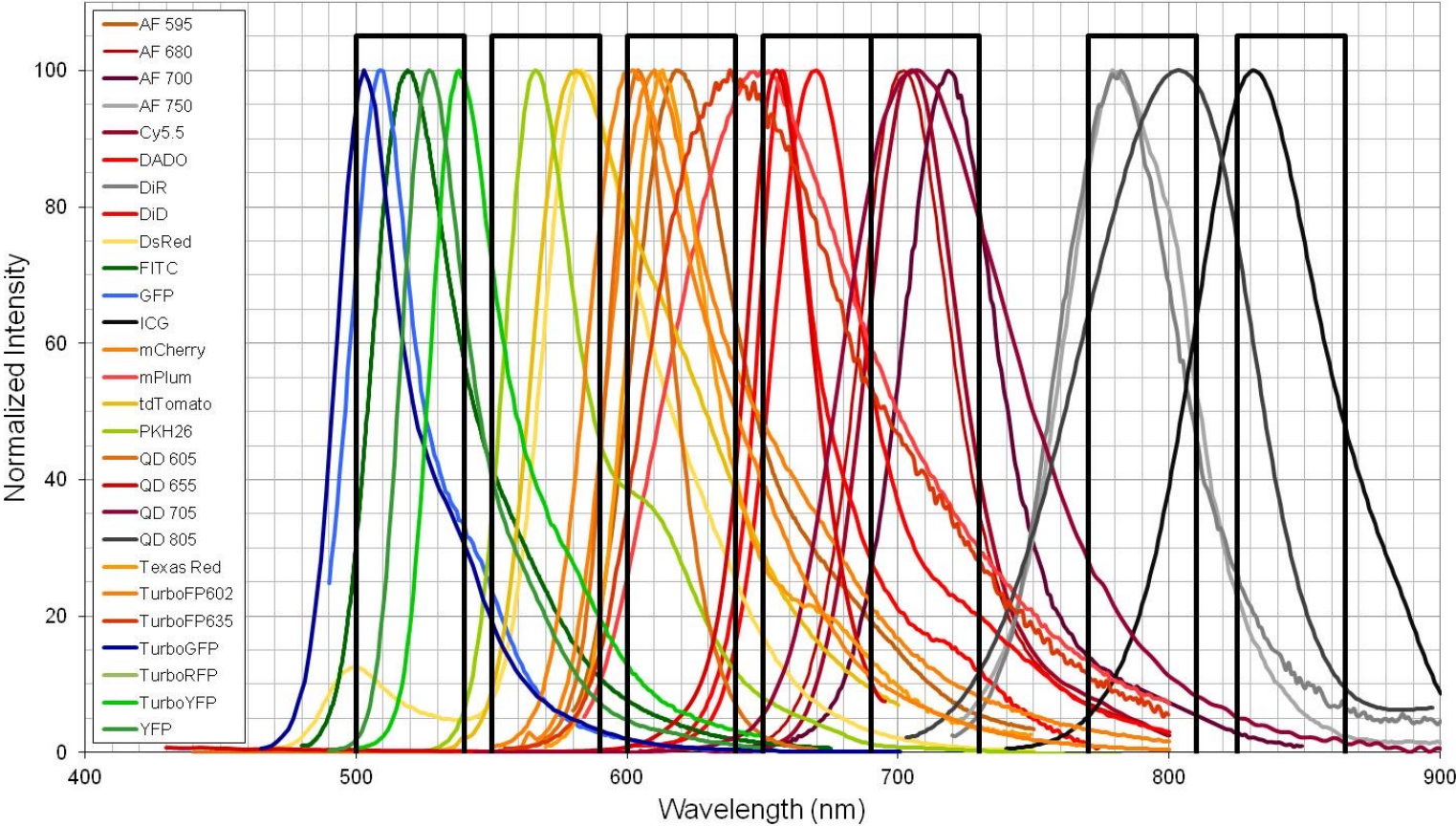


Narrow Bandwidth Excitation Filters Standard



Center Δ	Band pass
420	20
440	20
460	20
480	20
500	20
520	20
540	20
560	20
580	20
600	20
620	20
640	20
660	20
680	20
700	20
720	20
740	20
760	20
780	20

High Efficiency Emission Filters Centered Around Common Fluorophores



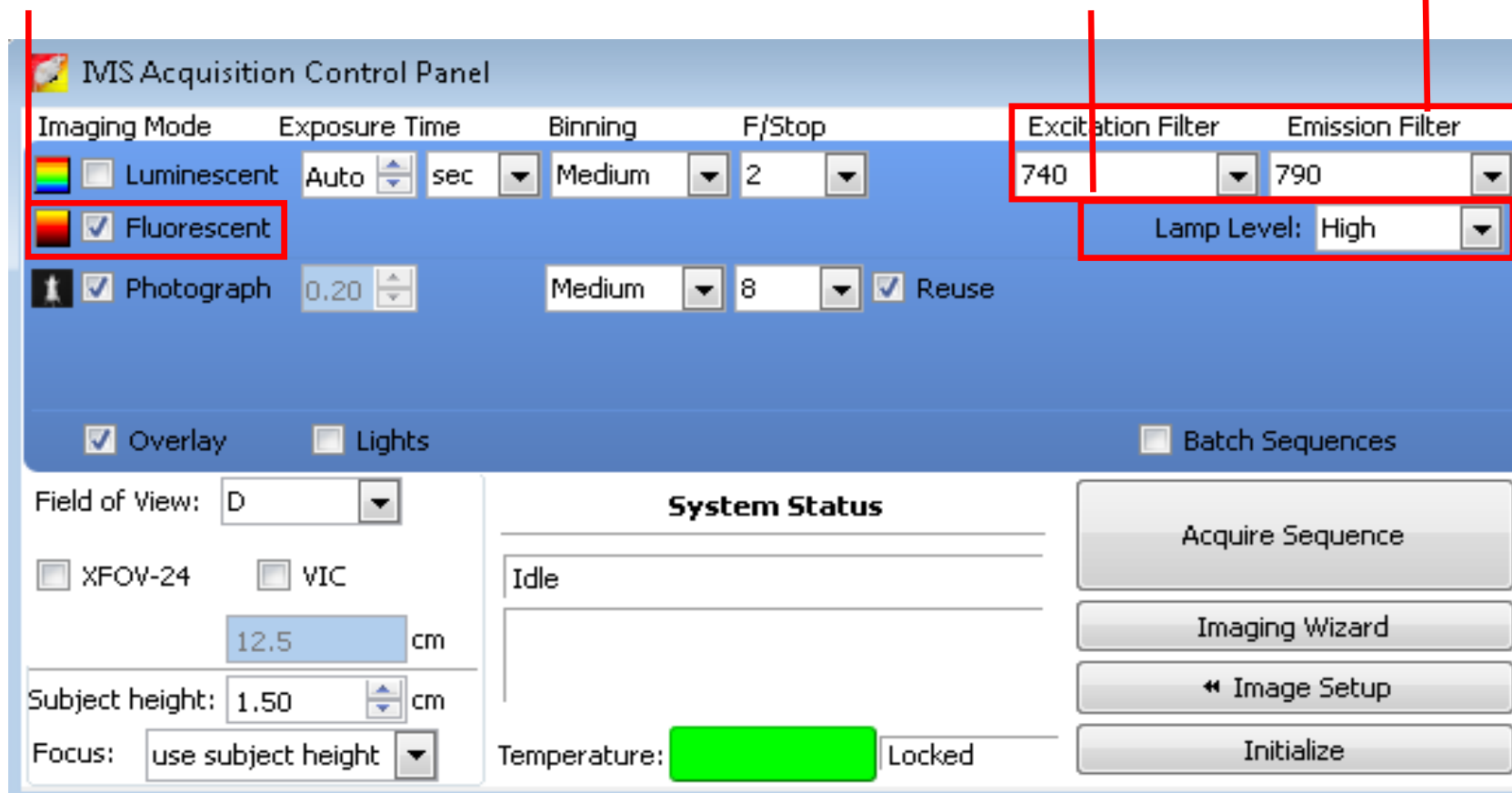
Center Δ	Band pass
520	40
570	40
620	40
670	40
710	40
790	40
845	40

Fluorescence Acquisition

Select Fluorescent
Imaging Mode

Lamp level High /
Low

Select filters



The screenshot shows the MIS Acquisition Control Panel interface. Red boxes and lines highlight specific settings:

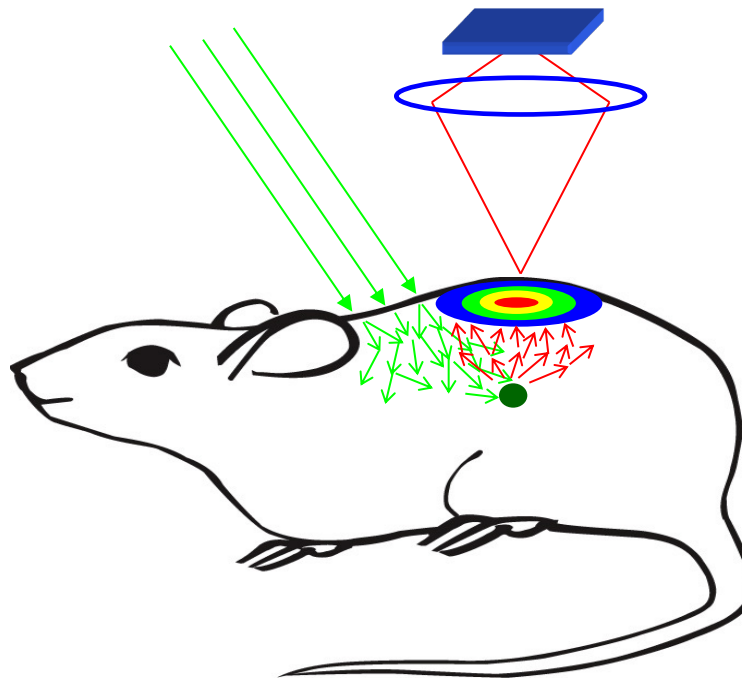
- Imaging Mode:** A red box highlights the **Fluorescent** mode, which is selected with a checkmark. The **Luminescent** mode is unselected.
- Excitation Filter:** A red box highlights the **Excitation Filter** dropdown menu, which is set to **740**.
- Emission Filter:** A red box highlights the **Emission Filter** dropdown menu, which is set to **790**.
- Lamp Level:** A red box highlights the **Lamp Level** dropdown menu, which is set to **High**.

Other visible settings include:

- Exposure Time:** Auto (sec)
- Binning:** Medium
- F/Stop:** 2
- Photograph:** 0.20 (Medium)
- Reuse:** Checked
- Overlay:** Checked
- Lights:** Unchecked
- Batch Sequences:** Unchecked
- Field of View:** D
- XFOV-24:** Unchecked
- VIC:** Unchecked
- Subject height:** 12.5 cm
- Subject height:** 1.50 cm
- Focus:** use subject height
- System Status:** Idle
- Temperature:** Locked
- Buttons:** Acquire Sequence, Imaging Wizard, Image Setup, Initialize

Fluorescent Calibrated Units: Radiant Efficiency

$$\text{Radiant Efficiency} = \frac{\text{Emission Light (photons/sec/cm}^2\text{/str)}}{\text{Excitation Light (}\mu\text{W/cm}^2\text{)}}$$



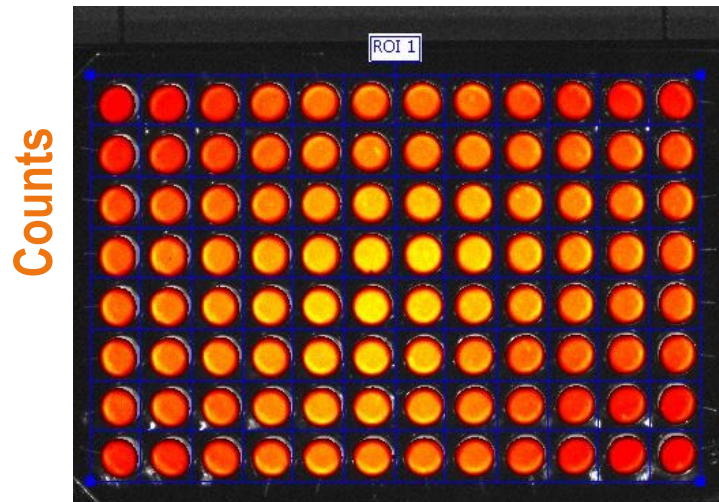
Fluorescent Calibrated Units: Radiant Efficiency

Excitation Light
Pattern



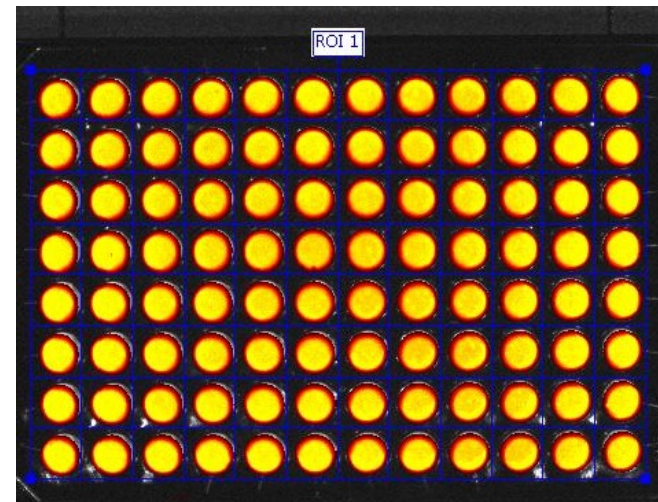
Units of 'Radiant Efficiency' compensates for non-uniform excitation light pattern

GFP Well Plate Uncorrected

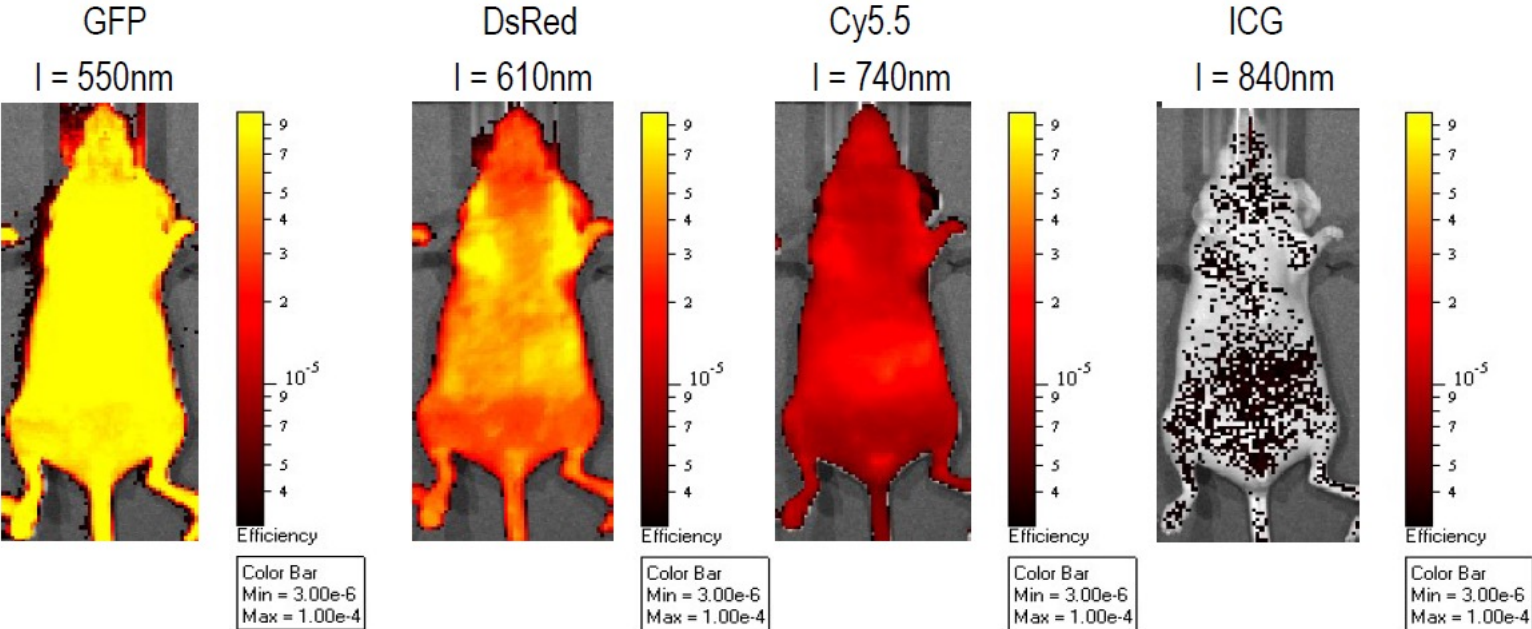


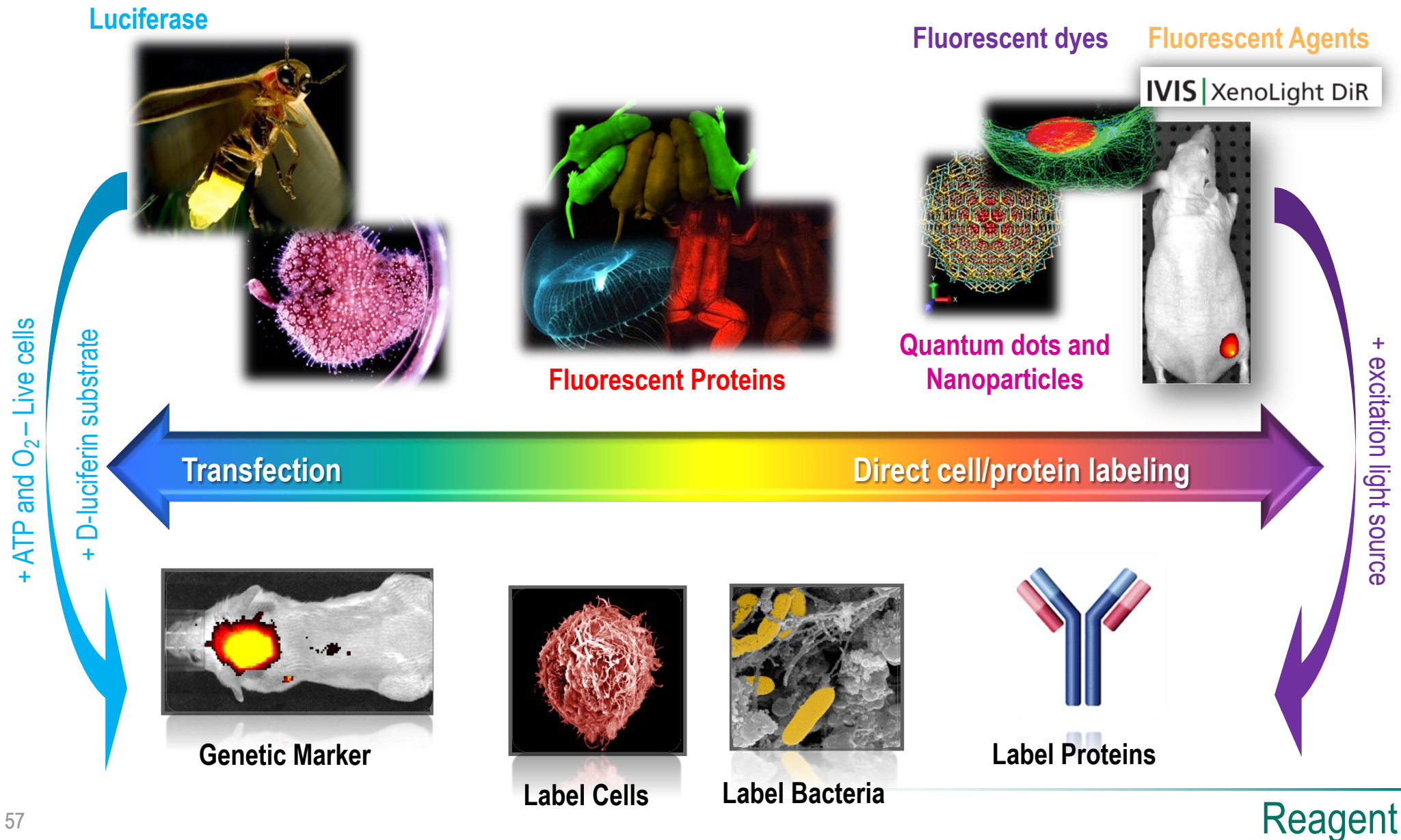
vs.

GFP Well Plate Corrected

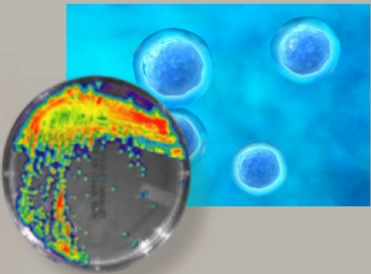


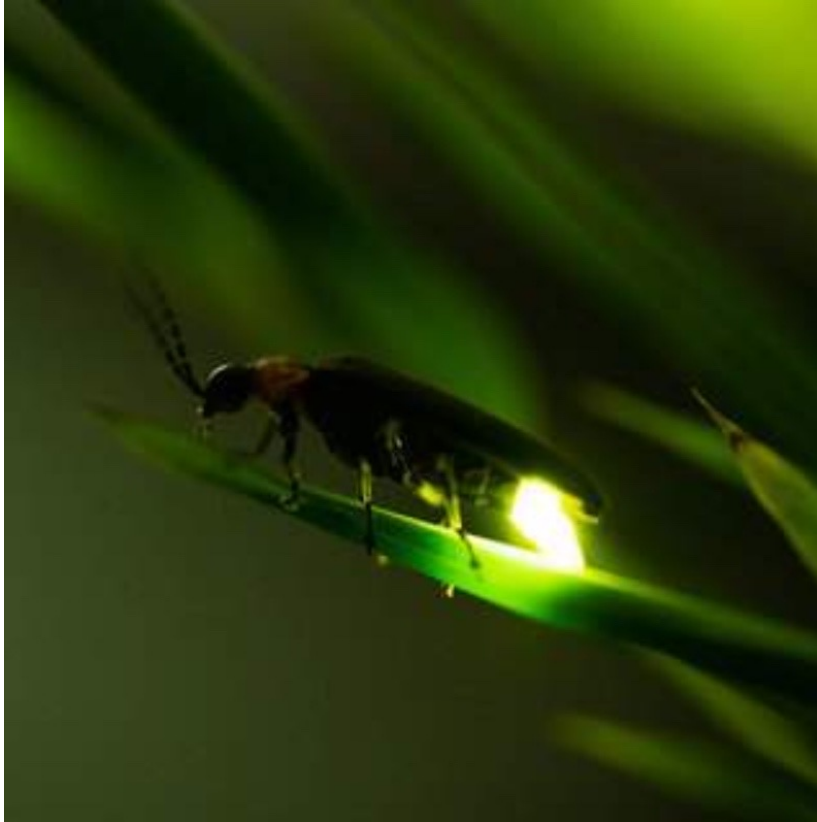
Radiant Efficiency



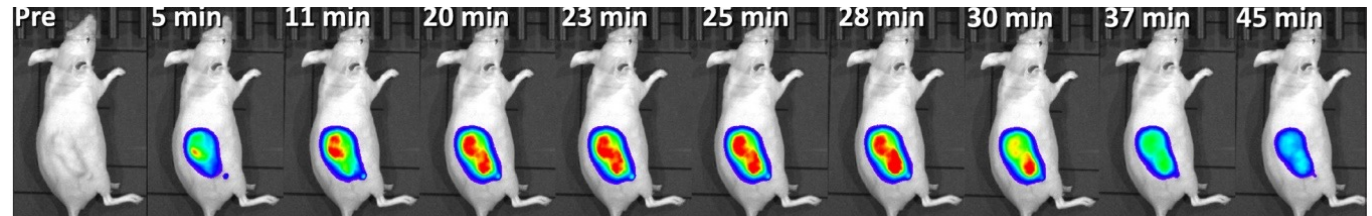


Optical Reagents

			
Bioluminescent Substrates	Bioluminescent Cells and Bacteria	Fluorescent Agents**	NIR Fluorescent Labels and Nanoparticles*
• Xenolight® D-Luciferin K+ Salt+ • XenoLight RediJect™ D-Luciferin • XenoLight RediJect D-Luciferin Ultra • XenoLight RediJect Ceolenterazine	• Bioware® Brite oncology cell lines with enhanced Red-Fluc vector • Bioware Brite Dual optical oncology cell lines with Red-Fluc and Green Fluorescent Protein (GFP) • RediFect Lentiviral Particles • Bacteria labeled with luciferase	• Imaging of a broad range of biomarkers and pathways • Activatable • Targeted • Vascular • Fluorescent Panels prepackaged and targeted for your research also available	• Labeling kits and dyes • Nanoparticles (645, 680, 750, 800 nm)



- ▶ RediJect™ solution – a pre-formulated, ready-to-use.
- ▶ Lyophilized D-Luciferin – offers the same sensitivity and high performance as the RediJect formulation.



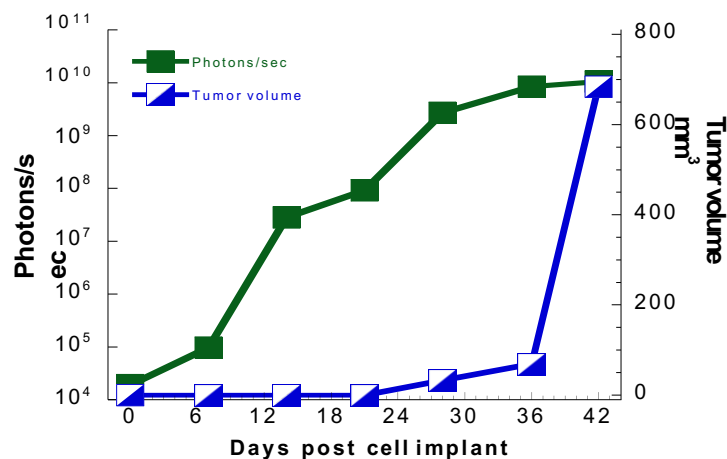
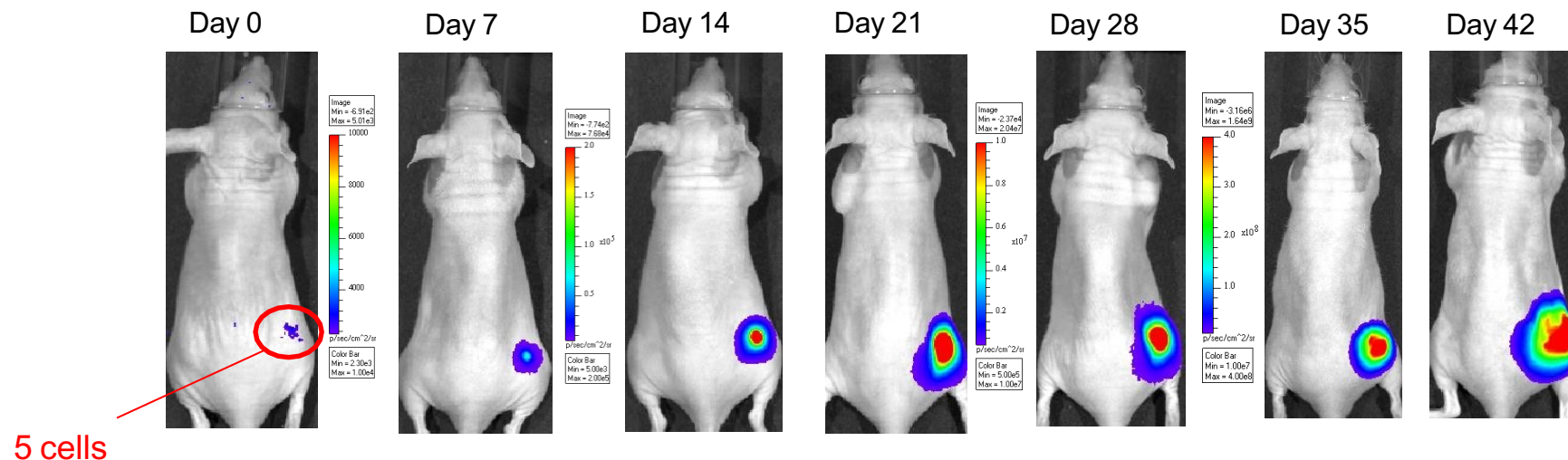
IVISbrite Bioluminescent Tumor Cell Lines

CAT. NUMBER	Name	Species	Cancer
BW134246	IVISbrite GL261 Red F-luc Bioluminescent Tumor Cell line	mouse	glioma
BW124577	IVISbrite U87MG Red F-luc Bioluminescent Tumor Cell line	human.	brain glioblastoma
BW124735	IVISbrite K562 Red F-luc Bioluminescent Tumor Cell line	human	chronic myelogenous leukemia
BW124353	IVISbrite HT-29 Red F-luc Bioluminescent Tumor Cell line	human	colorectal adenocarcinoma
BW124317	IVISbrite Colo205 Red F-luc Bioluminescent Tumor Cell line	human.	colorectal adenocarcinoma
BW124318	IVISbrite HCT-116 Red F-luc Bioluminescent Tumor Cell line	human	colorectal carcinoma
BW128092	IVISbrite HT1080 Red F-luc Bioluminescent Tumor Cell line	human	fibrosarcoma
BW124316	IVISbrite NCI-H460 Red F-luc Bioluminescent Tumor Cell line	human	lung adenocarcinoma
BW119266	IVISbrite A549 Red F-luc Bioluminescent Tumor Cell line	human	lung carcinoma
BW119267	IVISbrite LL/2 Red F-luc Bioluminescent Tumor Cell line	mouse	lung carcinoma

IVISbrite Bioluminescent Tumor Cell Lines

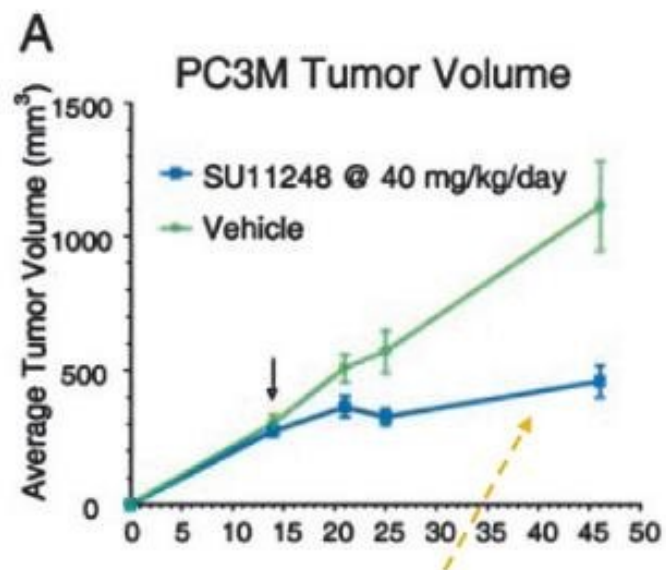
CAT. NUMBER	Name	Species	Cancer
BW134280	IVISbrite HepG2 Red F-luc Bioluminescent Tumor Cell line	human	hepatocellular carcinoma
BW119262	IVISbrite MCF-7 Red F-luc Bioluminescent Tumor Cell line	human	mammary adenocarcinoma
BW124087	IVISbrite 4T1 Red F-luc Bioluminescent Tumor Cell line	mouse	mammary gland adenocarcinoma
BW128090	IVISbrite 4T1 Red F-luc-GFP Bioluminescent Tumor Cell line	mouse	mammary gland adenocarcinoma
BW124734	IVISbrite B16F10 Red F-luc Bioluminescent Tumor Cell line	mouse	melanoma
BW119276	IVISbrite SKOV3 Red F-luc Bioluminescent Tumor Cell line	human	ovarian adenocarcinoma
BW125058	IVISbrite BxPC3 Red F-luc Bioluminescent Tumor Cell line	human	pancreatic adenocarcinoma
BW128444	IVISbrite PC3 Red F-luc Bioluminescent Tumor Cell line	human	prostate adenocarcinoma
BW133416	IVISbrite PC3 Red F-luc-GFP Bioluminescent Tumor Cell line	human	prostate adenocarcinoma
BW125055	IVISbrite LNCaP Red F-luc Bioluminescent Tumor Cell line	human	prostate carcinoma

Bioware Ultra: 4T1-luc2

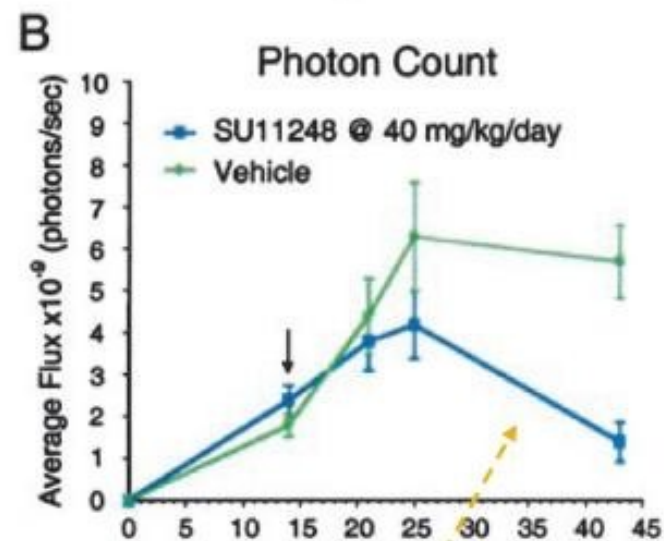


With Bioware Ultra one can start collecting data from Day 0, while with caliper measurements one has to wait at least 28 days to see any tumor growth

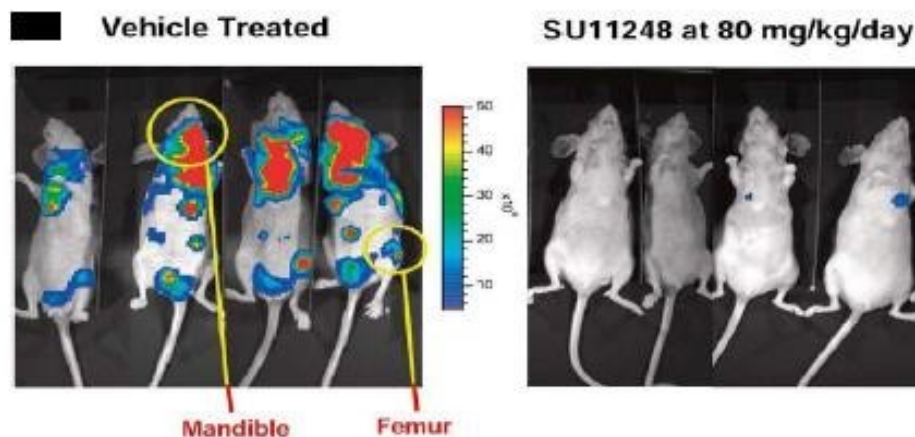
Sutent – Fast Tracked FDA Approval



Physical measurement
(tumor still getting bigger)

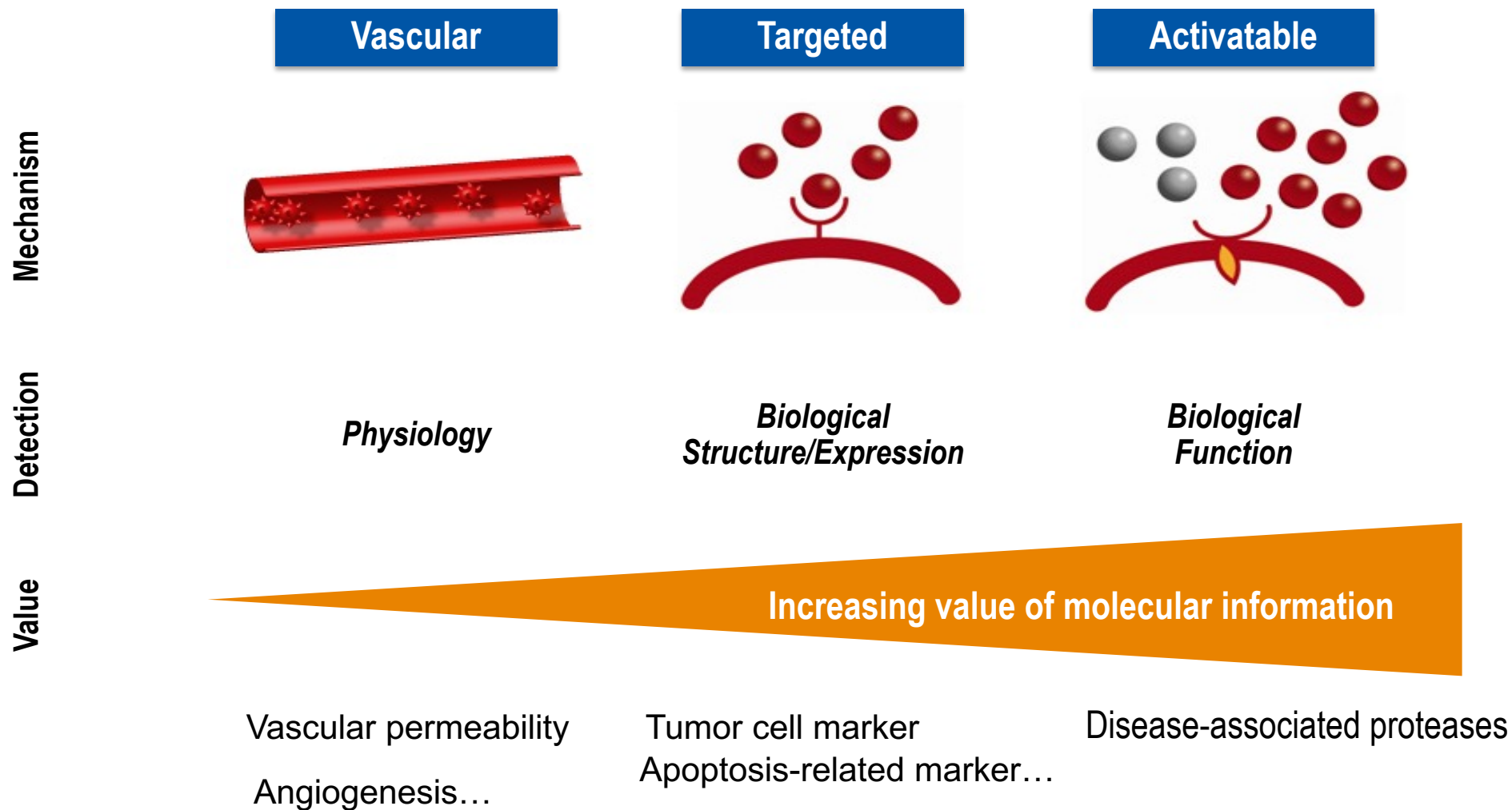


Biophotonic imaging
(tumor cells being killed)



Murray et al 2003

Agent Categories



PROBE	PRODUCT DESCRIPTION	CAT. NUMBER
IVISense Vascular 680	Imaging of vascularity, perfusion, and vascular permeability. Remains localized in vasculature for 0-4 h; accumulates in tumors and arthritic joints at 24 h.	NEV10054EX
IVISense Vascular 750		NEV10011EX
IVISense Vascular NP 680	Fluorescent nanoparticles for imaging vascularity, perfusion and vascular permeability; long pharmacokinetic profile.	NEV10149
IVISense Vascular NP 750		NEV10150
IVISense Gastrointestinal 750	Imaging to monitor gastric emptying and the impact of various drugs on gastric motility; may also be used as an anatomical marker for the stomach.	NEV11121
IVISense Acute Vascular 680 (1 mg)	Fluorescent probes that enable imaging and quantitation of acute vascular permeability increase, bladder clearance via intravital ureter imaging, or incorporation into liposomes for drug delivery monitoring.	NEV10117
IVISense Acute Vascular 680 (5 mg)		NEV10130
IVISense Acute Vascular 750 (1 mg)		NEV10118
IVISense Acute Vascular 750 (5 mg)		NEV10177

PROBE	PRODUCT DESCRIPTION	CAT. NUMBER
IVISense GFR 680	NIR-fluorescent imaging agent to non-invasively determine glomerular filtration rate (GFR) <i>in vivo</i> in models of kidney disease, dysfunction, and drug toxicity.	NEV30000
IVISense Edema 680	Blood pooling fluorescent probe for imaging circulation, blood vessels, vasculature, vascular leak, including that associated with early oncologic and ophthalmologic lesions. This agent has a short pharmacokinetic profile with bladder clearance, and binds to albumin in blood for a modestly extended (30m-1h) circulation half-life.	NEV10116



A range of highly fluorescent Physiologic Agents

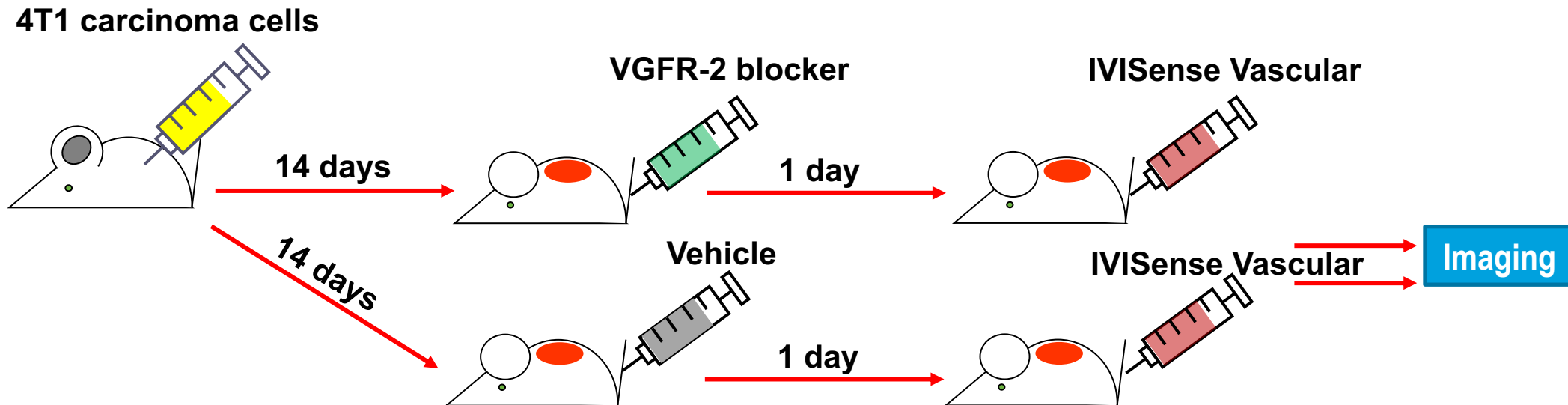
Remain stable and localized in the anatomy for various periods of time

Always fluorescent, circulate with blood or move through GI tract

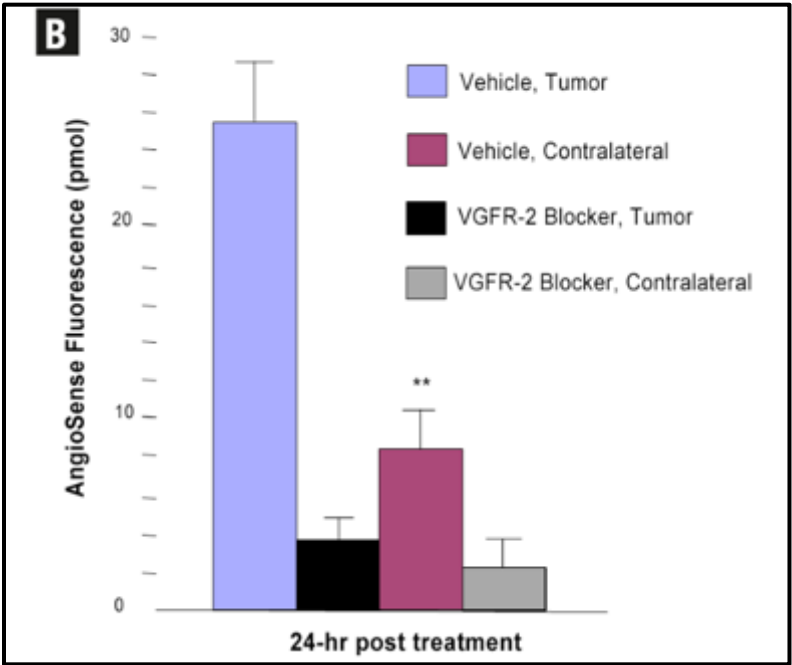
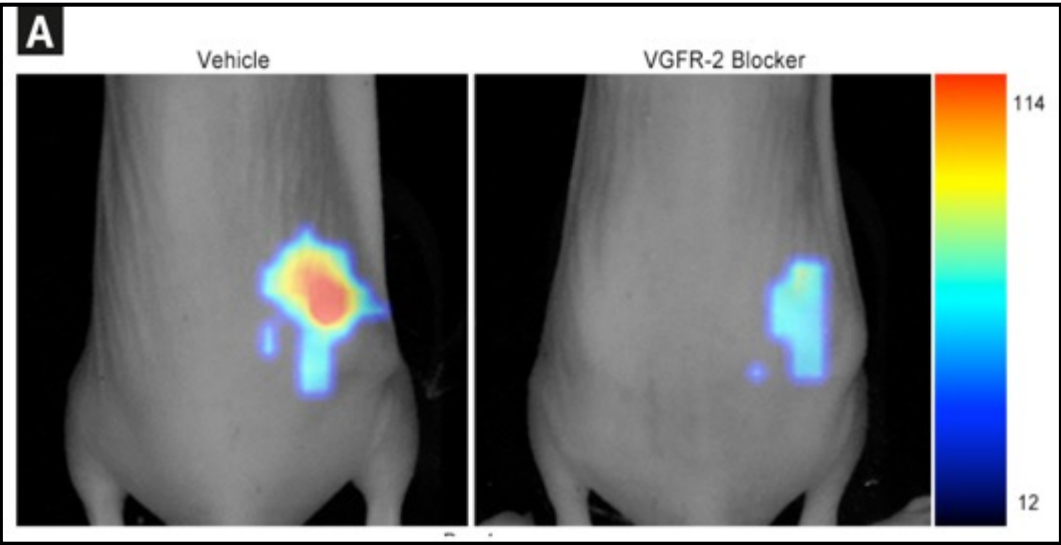
- Designed for in vivo use
- Limited in vitro applications

Monitor the integrity of the vascular system

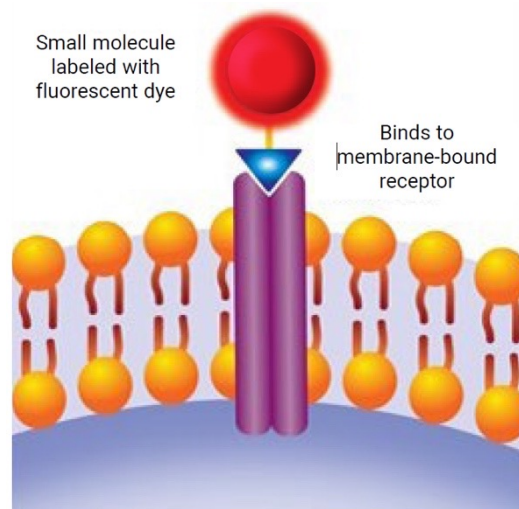
Monitoring Tumor Development in a Mammary Fat Pad Model Using Implanted 4T1 Tumor Cells



Monitoring Tumor Development in a Mammary Fat Pad Model Using Implanted 4T1 Tumor Cells



IVISense Fluorescent Probes - Targeted



PROBE	PRODUCT DESCRIPTION	CAT. NUMBER
IVISense Annexin-V 750	<i>In vivo</i> targeting of membrane-bound phosphatidylserine exposed during the early stages of apoptosis.	NEV11053
IVISense Bombesin Receptor 680	Target and quantify up-regulation of bombesin receptors (BBR) <i>in vivo</i> associated with tumor proliferation. These receptors are also over-expressed in a variety of cancers.	NEV10090
IVISense Folate Receptor 680	Highly specific and sensitive in the detection of folate receptor proteins. Can be used to closely monitor and quantitate tumor growth and metabolism.	NEV10040
IVISense Hypoxia CA IX 680	Detects the tumor cell surface expression of carbonic anhydrase 9 (CA IX) protein, which increases in hypoxic regions within many tumors.	NEV11070
IVISense Integrin Receptor 645	Targets integrin $\alpha v \beta 3$ expressed in oncology, atherosclerosis, and angiogenesis disease models.	NEV10640
IVISense Integrin Receptor 680		NEV10645
IVISense Integrin Receptor 750		NEV10873

PROBE	PRODUCT DESCRIPTION	CAT. NUMBER
IVISense Osteo 680	Optimized imaging of bone turnover through binding of hydroxyapatite.	NEV10020EX
IVISense Osteo 750		NEV10053EX
IVISense Osteo 800		NEV11105
IVISense Tomato Lectin 680	NIR-labeled tomato lectin protein that has high binding affinity for glycoprotein N-acetylglucosamines on the surface of vascular endothelial cells. Use for vascular mapping <i>in vivo</i> .	NEV10060
IVISense Transferrin Receptor 750	NIR-labeled transferrin detects transferrin receptor upregulation associated with the increased cell metabolic need for iron in cancer and inflammatory cells. Also detects normal iron metabolism in the liver.	NEV10091
IVISense Bacterial 750 Probe in RediJect™ Solution (1 vial)	NIR-targeted probe for non-invasive detection of bacterial infections <i>in vivo</i> .	133397
IVISense Bacterial Probe 750 in RediJect Solution (4 vials)		133398
IVISense Bacterial 750 Control Dye in RediJect Solution (1 vial)	Non-reactive control dye for RediJect Bacterial Detection Probe.	133399
IVISense 2-DG 750 Probe in RediJect Solution (1 vial)	NIR-targeted probe for non-invasive imaging of glucose uptake <i>in vivo</i> .	760561
IVISense 2-DG 750 Probe in RediJect Solution (4 vials)		760562
IVISense 2-DG 750 Control Dye in RediJect Solution (1 vial)	Non-reactive control dye for RediJect 2-DG 750 probe.	760567

Targeted Agents (IVISense Annexin-V 750)

Targets **p**hosphatidyl**s**erine (**PS**) exposed in the outer leaflet of the lipid bilayer during the **early stages of apoptosis**

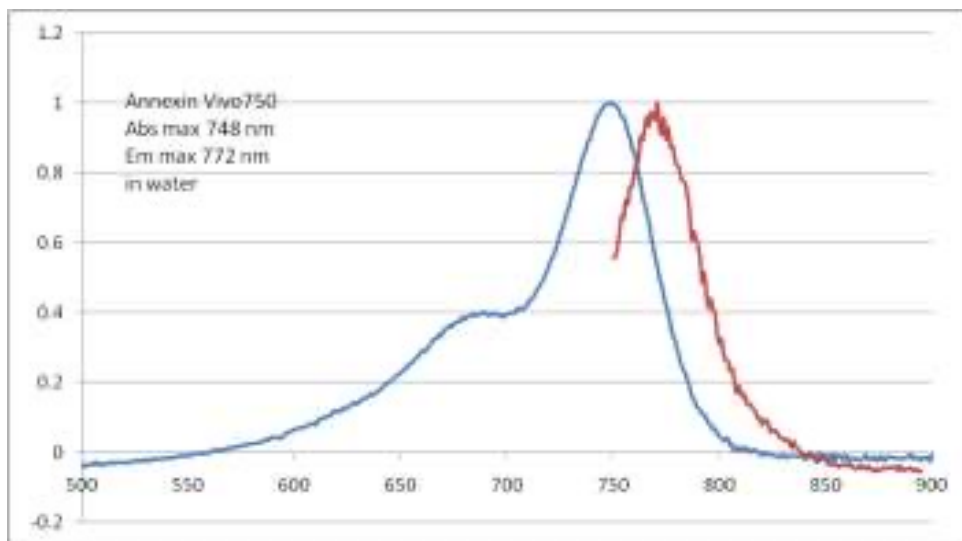
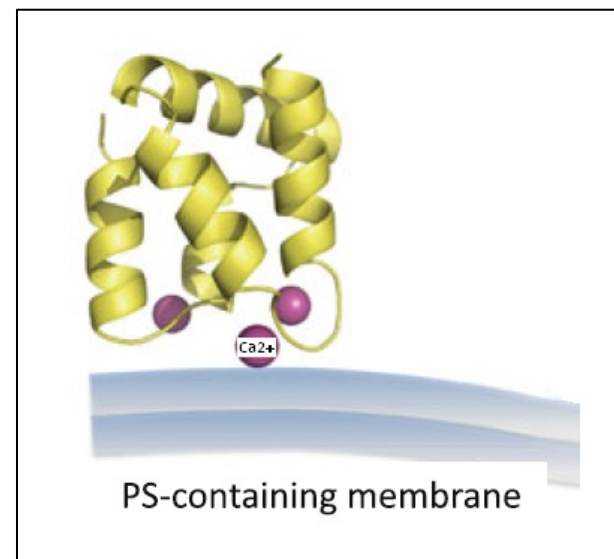


Figure 1. Absorbance and fluorescence emission spectra of Annexin-Vivo 750 in 1x PBS.

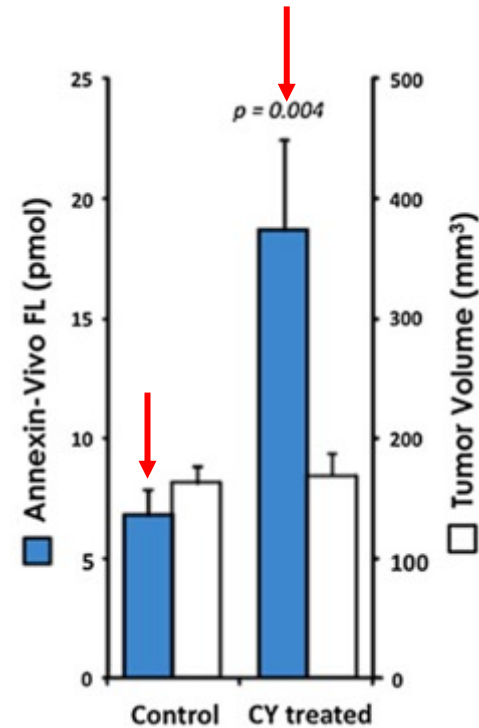
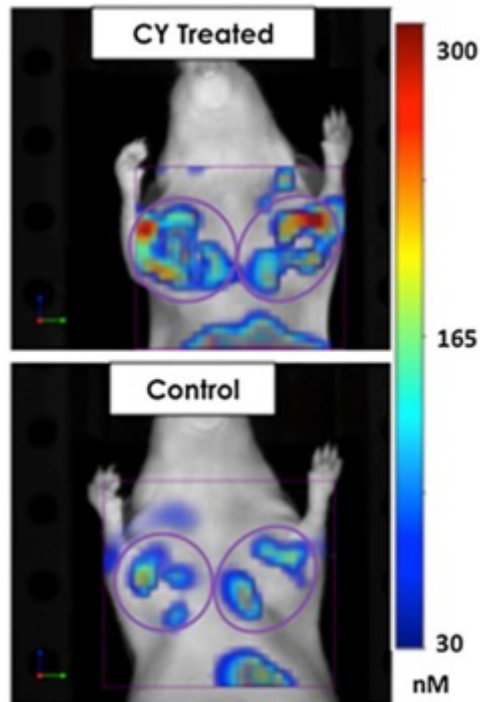
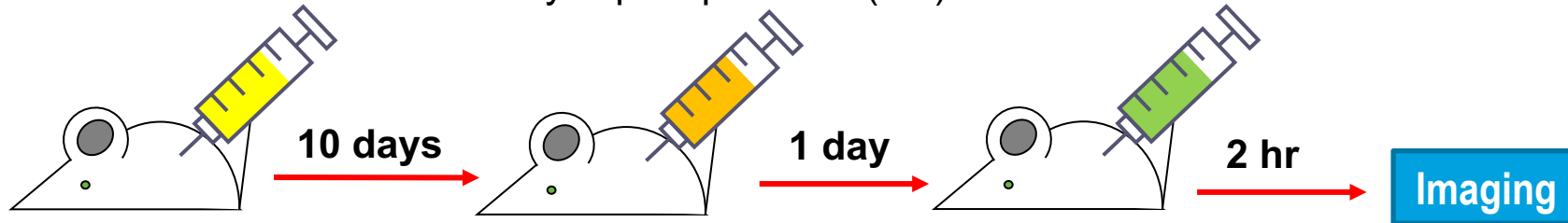


Case Study- Anti-cancer treatment (IVISense Annexin-V 750)

HT-29 human colorectal
adenocarcinoma cells

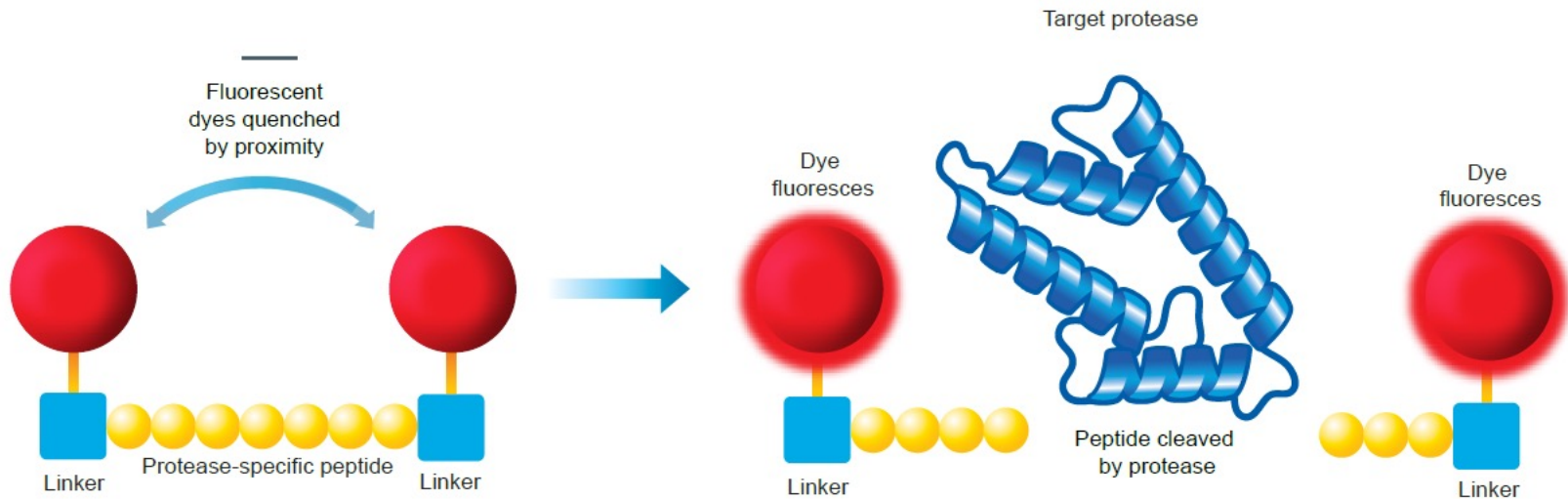
Cyclophosphamide (CY)

IVISense Annexin-V 750



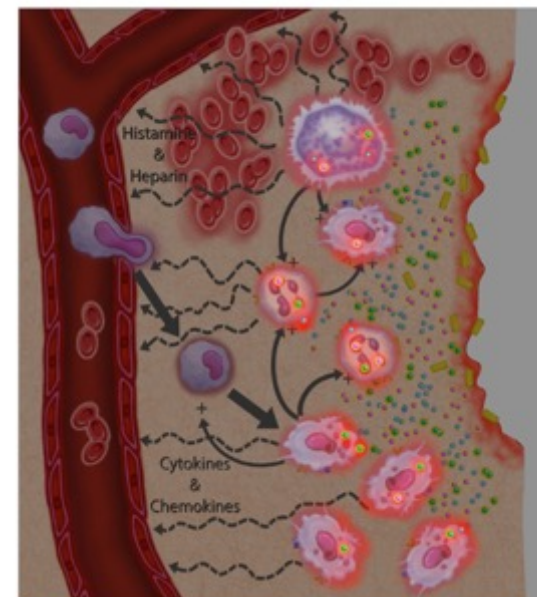
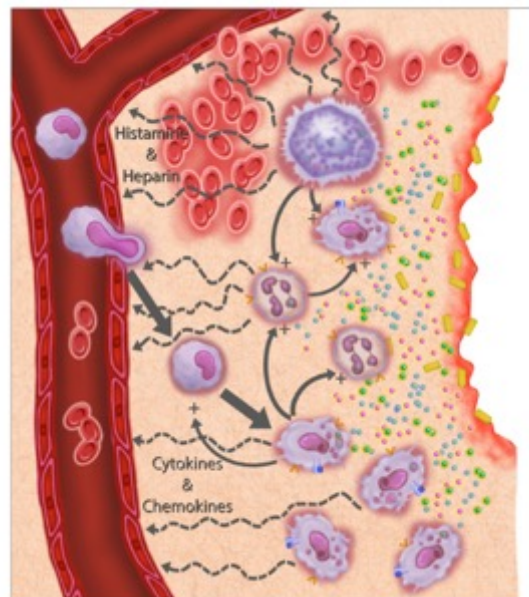
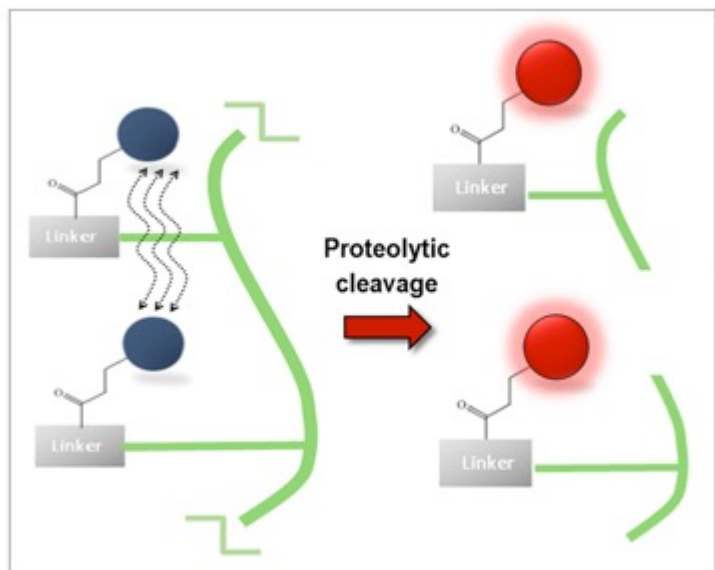
PROBE	PRODUCT DESCRIPTION	CAT. NUMBER
IVISense Cat B 680 FAST	Selective imaging of cathepsin B proteinases (Cat B). Optically silent in the inactivated state, becoming highly fluorescent when activated.	NEV11112
IVISense Cat B 750 FAST		NEV11098
IVISense Cat K 680 FAST	Imaging of cathepsin K (Cat K) activity in oncology applications involving metastasis to the bone as well as a broad range of bone applications, including bone loss, tumor-induced osteolysis, and bone changes following arthritis.	NEV11000
IVISense MMP 645 FAST	Imaging of metalloproteinase (MMP) activity is involved in many disease-related phenomena, including cancer propagation, invasion and metastasis, rheumatoid arthritis, and areas of cardiovascular disease.	NEV10100
IVISense MMP 680		NEV10126
IVISense MMP 750 FAST		NEV10168

PROBE	PRODUCT DESCRIPTION	CAT. NUMBER
IVISense Neutrophil Elastase 680 FAST	Fluorescent neutrophil elastase-activatable agent that is optically silent upon injection and produces a fluorescent signal after cleavage by elastase produced by neutrophil cells.	NEV11169
IVISense Pan Cathepsin 680 (formerly ProSense 680)	Versatile imaging of changes in cathepsin-based protease activity as seen in a number of pathological states and disease-related events, including rheumatoid arthritis, cancer, atherosclerosis, angiogenesis, and cardiovascular disease.	NEV10003
IVISense Pan Cathepsin 750		NEV10001EX
IVISense Pan Cathepsin 750 FAST	FAST version of ProSense, with faster kinetics and a broader imaging window.	NEV11171
IVISense Renin Receptor 680 FAST	Imaging of renin-angiotensin pathway associated with hypertension, and kidney and cardiovascular disease.	NEV11079



Activatable Agents (IVISense Pan Cathepsin 680)

- A protease activatable fluorescent *in vivo* imaging agent.
- Activated by key disease associated proteases such as Cathepsin B, L, S and Plasmin.
- For optimal cancer and inflammation-related cathepsin visualization.

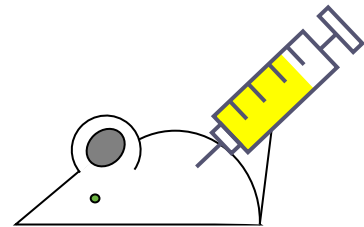


Case Study- Asthma (IVISense Pan Cathepsin 680)

Induce allergic reaction

Intranasal challenge

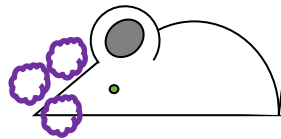
Ovalbumin (OVA)



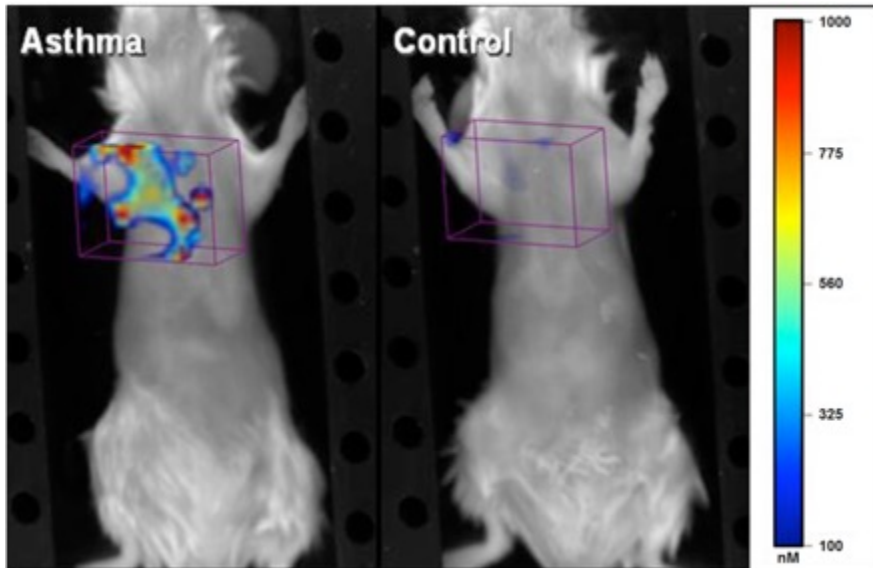
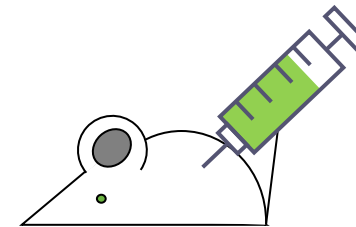
3 weeks



OVA



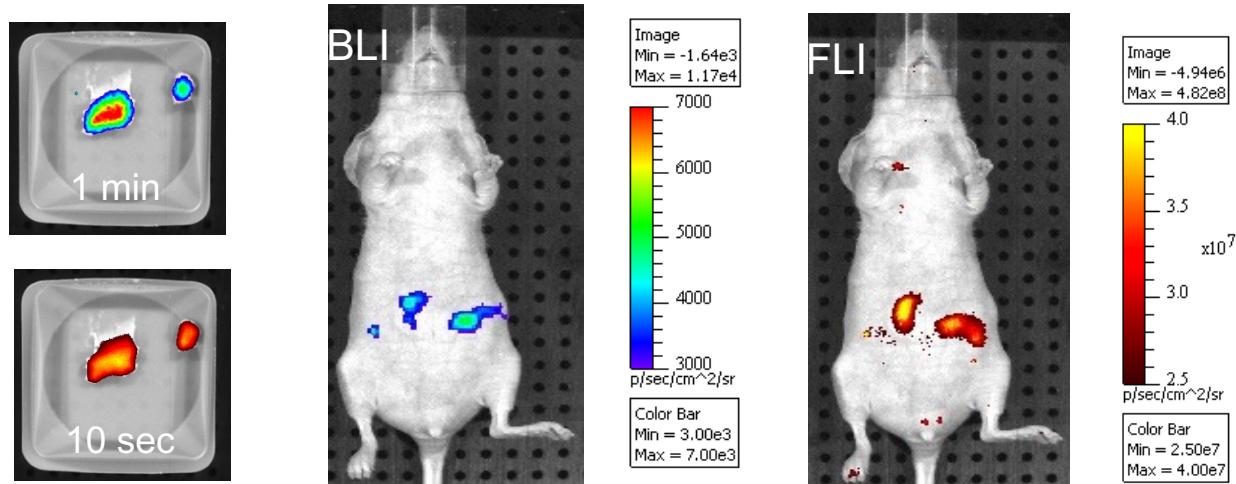
IVISense Pan Cathepsin 680



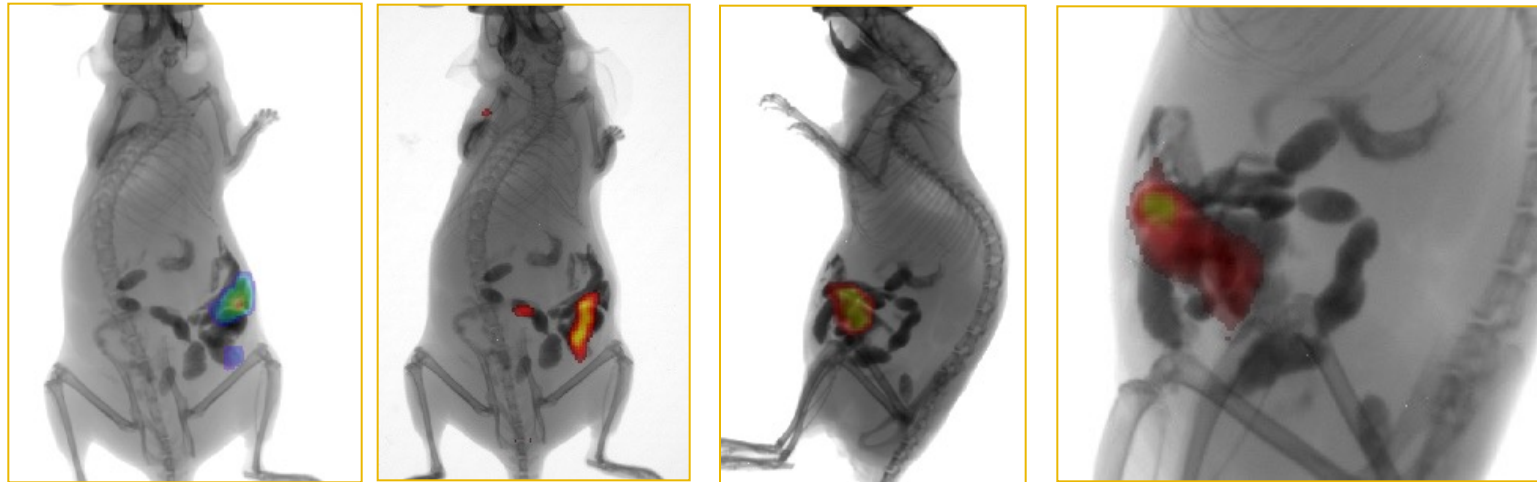
Asthmatic mouse (left) injected IVISense Pan Cathepsin 680, shows broad distribution of fluorescence and inflammation in the lung.

Gastrointestinal tract infection model

A.



B.



A GI tract infection model was established by feeding a mouse with contaminated peanut butter, which contained bioluminescence and fluorescence dually labeled *Salmonella typhimurium* (Xen26-lux-cherry). Bioluminescence and fluorescence (Ex605/Em660 nm) images were taken at 3 hours (A). At 5 hours, tri-modality imaging was performed and the overlaid images were shown (B). The GI tract was highlighted due to the presence of barium sulfate (150 mg) in the peanut butter.

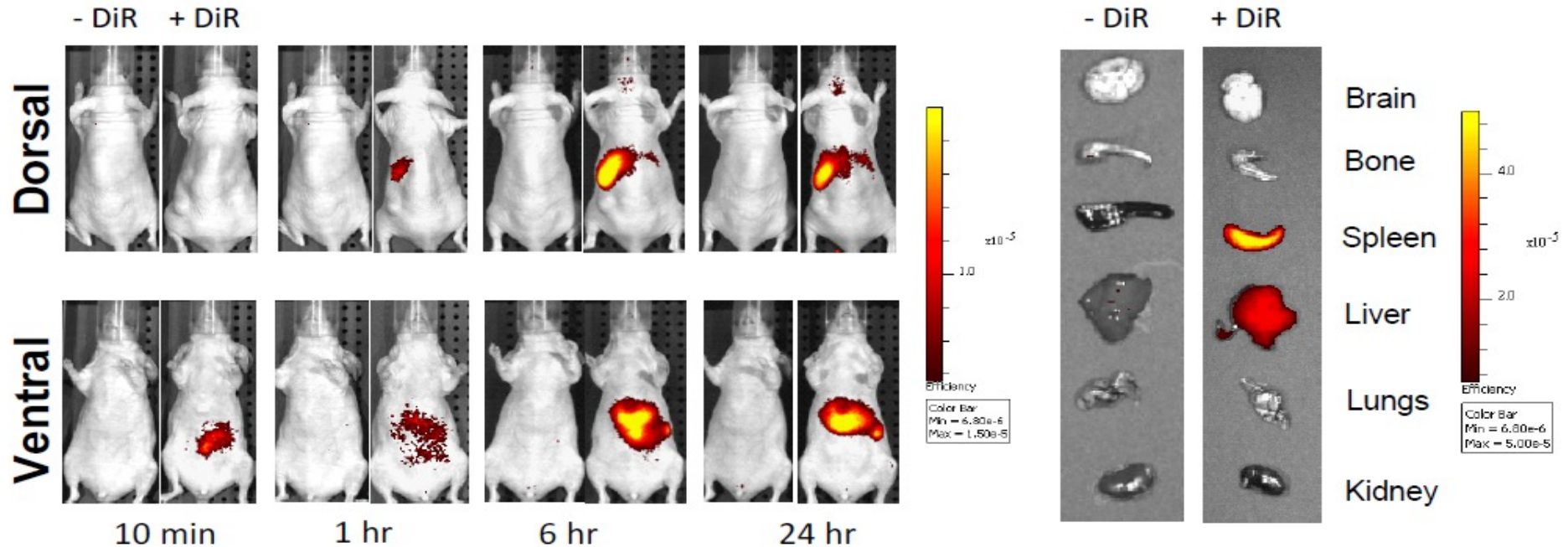
PerkinElmer offers four categories of fluorescent *in vivo* imaging agents:

LABELS and NANOPARTICLES

VivoTag™ 680XL Protein Labeling Kit : designed for preparing fluorescently labeled antibodies, proteins or peptides for small animal in vivo imaging applications.

VivoTrack 680 : cell labeling agent that intercalates into the plasma membrane of primary cells and cell lines.

► *In Vivo* Imaging of DiR Stained Spleen T-cell Distribution



T-cells isolated from the spleen were fluorescently stained with DiR and i.v. injected (5×10^6 cells/mouse) into a Nu/Nu mouse. Images above taken 24hrs post injection with IVIS Spectrum show cells homing to the spleen

1. Choose reporters that maximize signal-to-noise (S:N) ratio
2. Consider the appropriate control groups and imaging time points necessary
3. Use hairless mice or white-furred animals and depilate or shave
4. Switch to autofluorescence-free mouse diet
5. Closely map the kinetics of your biological bioluminescent model
6. Animal handling can significantly affect kinetics
7. Image in the animal orientation that yields the highest signal intensity
8. Cover intense signal to allow dimmer signals to dictate auto-exposure
9. Utilize guards to prevent reflection off neighboring animals
10. Use black well plates when doing in vitro experimentation



HUMAN HEALTH | ENVIRONMENTAL HEALTH

Thanks for your attention!

J & H 博克科技股份有限公司

服務專線：0800-898-178

技術支援：support@jnhtech.com.tw